MACHINE LEARNING / MODULE 5

Rare events, anomalies, and model selection

Finding the unusual, judging rare-event evidence, and choosing models without fooling yourself

Full day · 09:00–17:00 | 7 parts | 2 laboratories | Course synthesis

Machine learning for energy applications

Today's plan

09:00	Welcome: rare events in energy operations	Lecture
09:15	Part 1: What counts as an anomaly	Lecture
09:55	Part 2: Detectors that learn normal behaviour	Lecture
10:45	Break	Break
11:00	Lab 1: statistical rules and Isolation Forest	Lab
12:05	Part 3: Evaluating rare events	Lecture
12:45	Lunch	Lunch
13:30	Part 4: Imbalance, calibration, uncertainty	Lecture
14:10	Lab 2: detector comparison and an alert policy	Lab
15:25	Break	Break
15:40	Part 5: Honest and efficient model selection	Lecture
16:05	Exercise: read a tuning log and choose a model	Exercise
16:20	Part 6: Deployment, drift, and monitoring	Lecture
16:40	Part 7: Course synthesis and exit check	Discussion

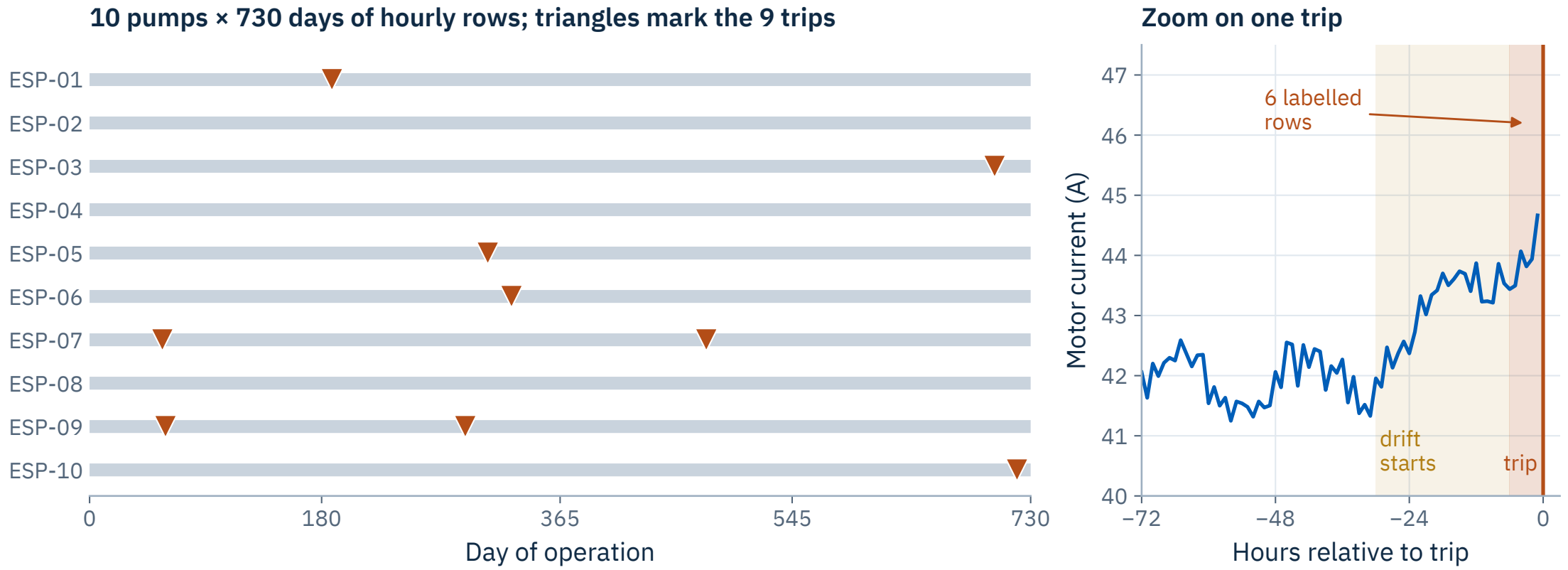
What you will be able to do by 17:00

By the end of the day you will be able to:

1. Match an anomaly problem to its setting and to its type: point, contextual, or collective.
2. Build robust statistical baselines and compare Isolation Forest, LOF, One-Class SVM, and reconstruction error.
3. Evaluate rare-event detectors with precision-recall, event-level metrics, calibration, and honest intervals.
4. Set and document an alert threshold from costs or from review capacity.
5. Run a bounded, leakage-free tuning experiment and compare models with nested CV and paired folds.
6. Plan drift monitoring and connect the full workflow of Modules 1 to 5.

Nine pump trips hide in 175,200 hourly rows

Labelling the 6 hours before each trip gives 54 positive rows: 0.03% of the data.



Synthetic fleet (seed 5): 10 pumps, 730 days, 9 trips. The zoomed current trace is illustrative, not field data.

PRACTICE

3 min · Pairs

Warm-up: judge a detector that never alerts

Same fleet: 175,200 hourly rows, of which 54 are labelled positive (6 per trip).

1. What accuracy does a model that never alerts achieve?
2. How many of the nine trips does it catch?
3. Which two numbers would you ask for instead?

PART 1 OF 7

01

What counts as an anomaly

Unusual is defined by a reference, a context, and a purpose.

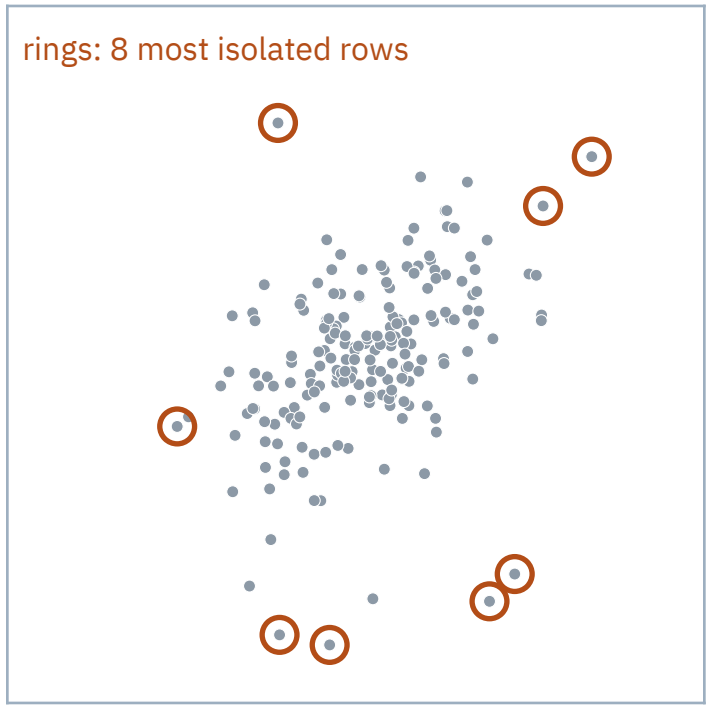
- Three meanings of anomaly and the labels behind them
- Point, contextual, and collective anomalies
- Robust statistical baselines and how they fail
- Physical rules that statistics cannot replace

Anomaly has several meanings

Setting	Training information	Typical question
Outlier detection	Mixed, mostly ordinary observations	Which training observations differ?
Novelty detection	A reference population intended to be normal	Does a new observation depart from normal?
Supervised rare-event prediction	Examples labelled by event outcome	How likely is the labelled event?

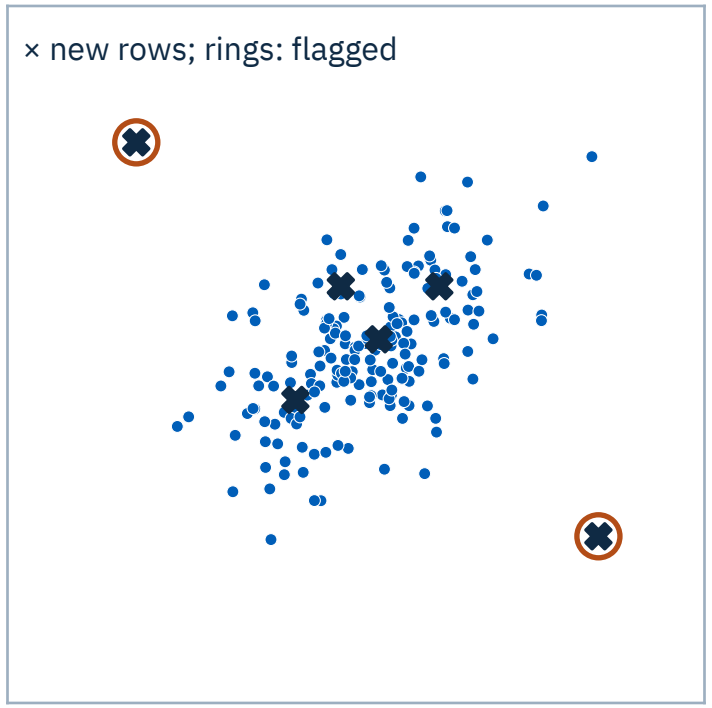
Available labels decide the learning setting

Outlier detection



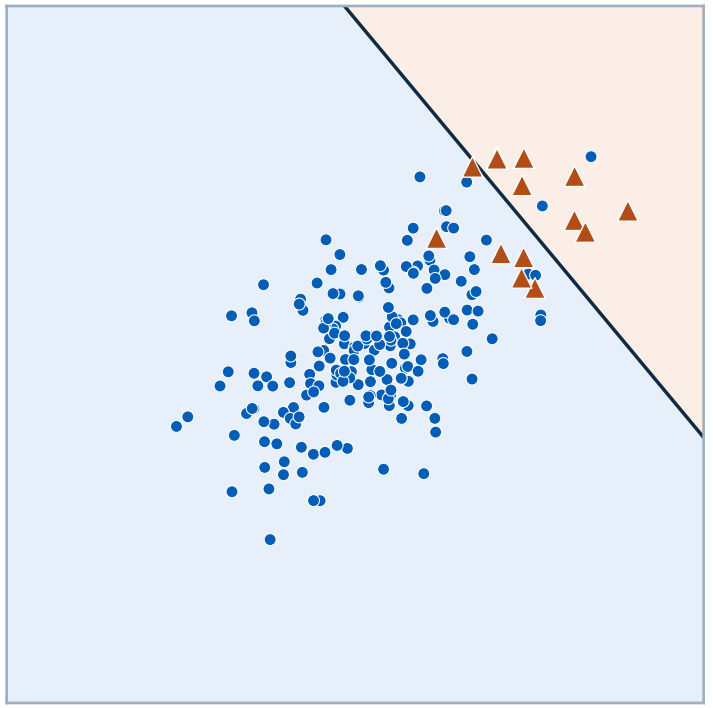
Training labels: none
(the data are contaminated)

Novelty detection



Training labels: normal only
(new rows scored later)

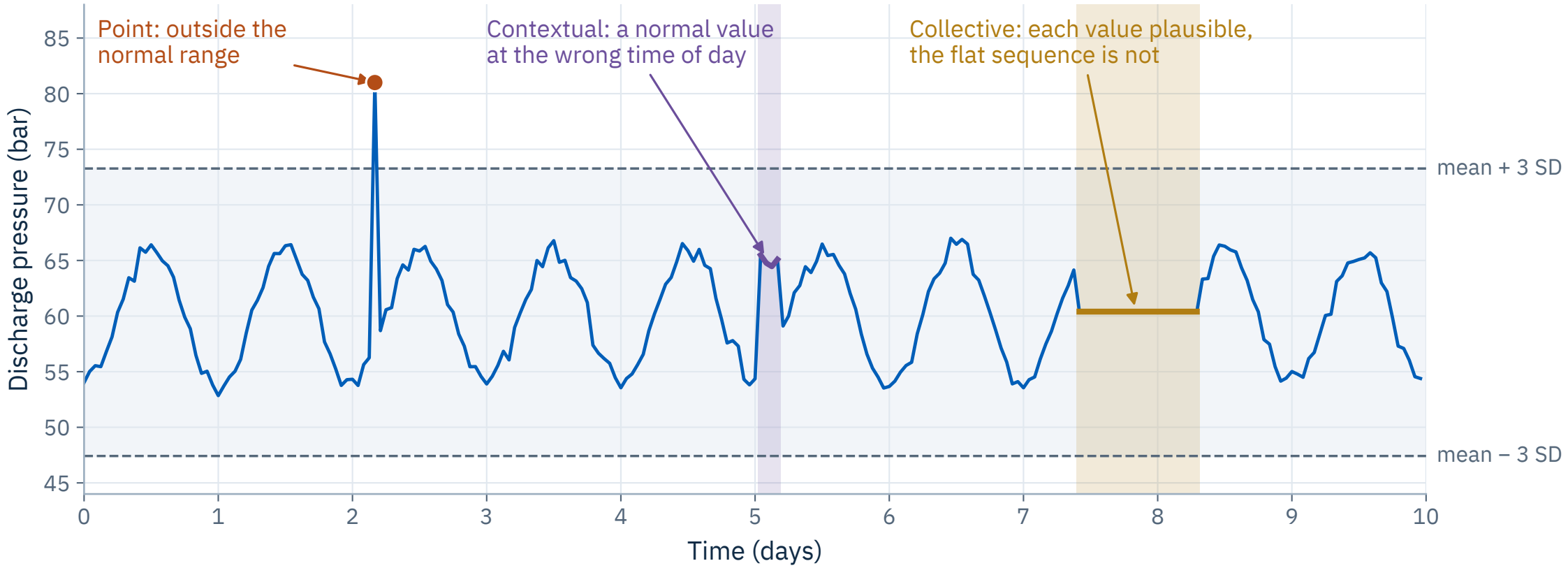
Supervised rare events



Training labels: normal and events
(14 events among 204 rows)

Synthetic 2-D data (seed 7). Left: Isolation Forest ranks a contaminated set. Middle: novelty LOF scores new rows against a clean reference. Right: logistic regression on labelled events.

Anomalies come as points, contexts, or collectives



Synthetic hourly discharge pressure (seed 11) with an injected spike, four peak-level hours at trough time, and a 22-hour flat segment.

Unusual is not the same as faulty

Rare is a property of the data; faulty is a property of the equipment.

What an anomaly score says

- This row is rare relative to the reference data
- A ranking under one model's idea of normal
- Depends on features, scaling, and settings

vs

What a fault diagnosis needs

- A physical mechanism and an engineering review
- A confirmed outcome, with its timing
- Evidence that the pattern recurs across assets

◆ Report detections as candidates for review, not as faults.

Statistical rules provide a transparent baseline

Start every anomaly project with rules you can explain line by line:

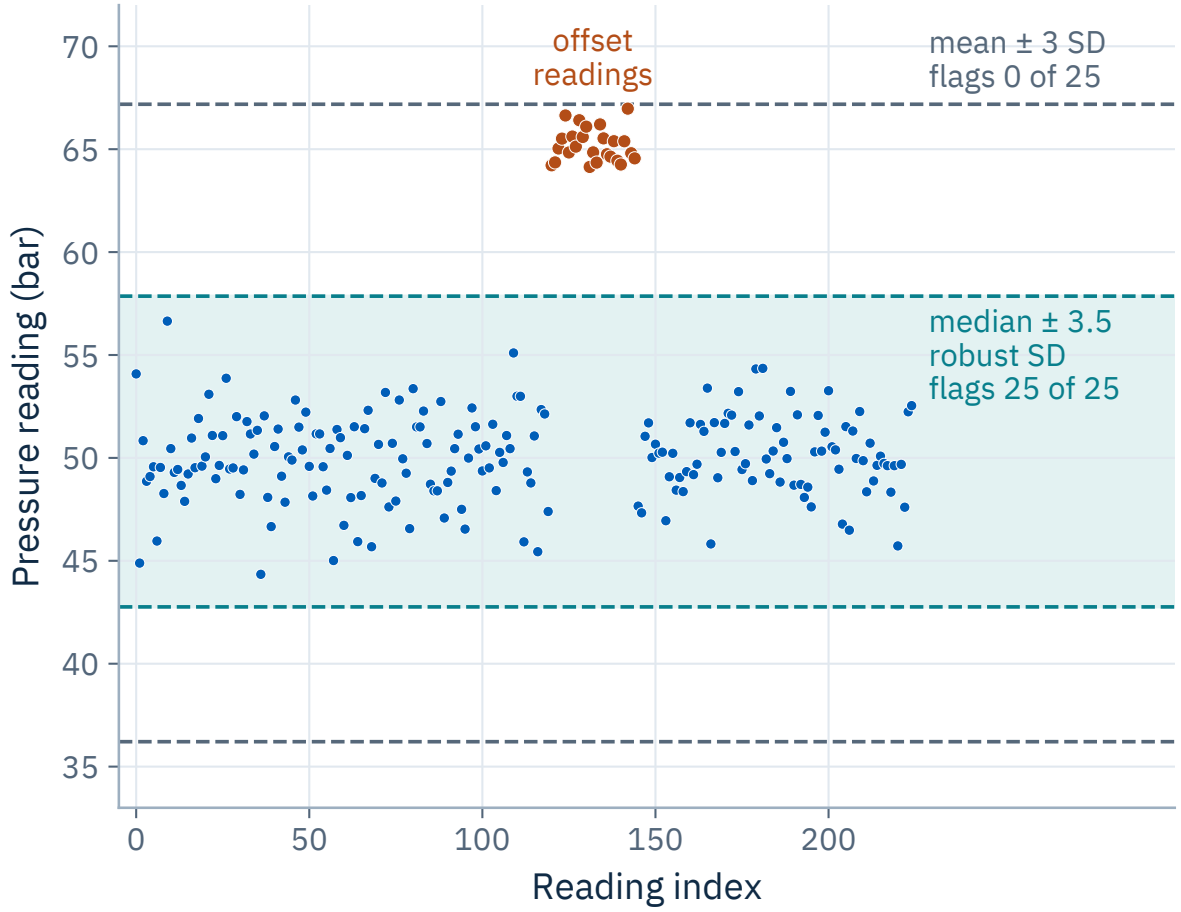
1. **Plot first** — each variable over time or depth, coloured by operating regime
2. **Describe robustly** — median and MAD per regime, not one global mean and SD
3. **Write physical rules** — ranges, rates of change, flatlines, channel consistency
4. **Flag and rank** — robust z beyond 3.5, fences per facies, rule violations
5. **Review before naming faults** — an engineer labels flagged and unflagged samples

◆ Treat every rule as an explicit, reviewable assumption.

Outliers inflate the statistics meant to catch them

$$z = \frac{x - \bar{x}}{s} \qquad z_{\text{robust}} = \frac{x - \text{median}}{1.4826\,\text{MAD}}$$

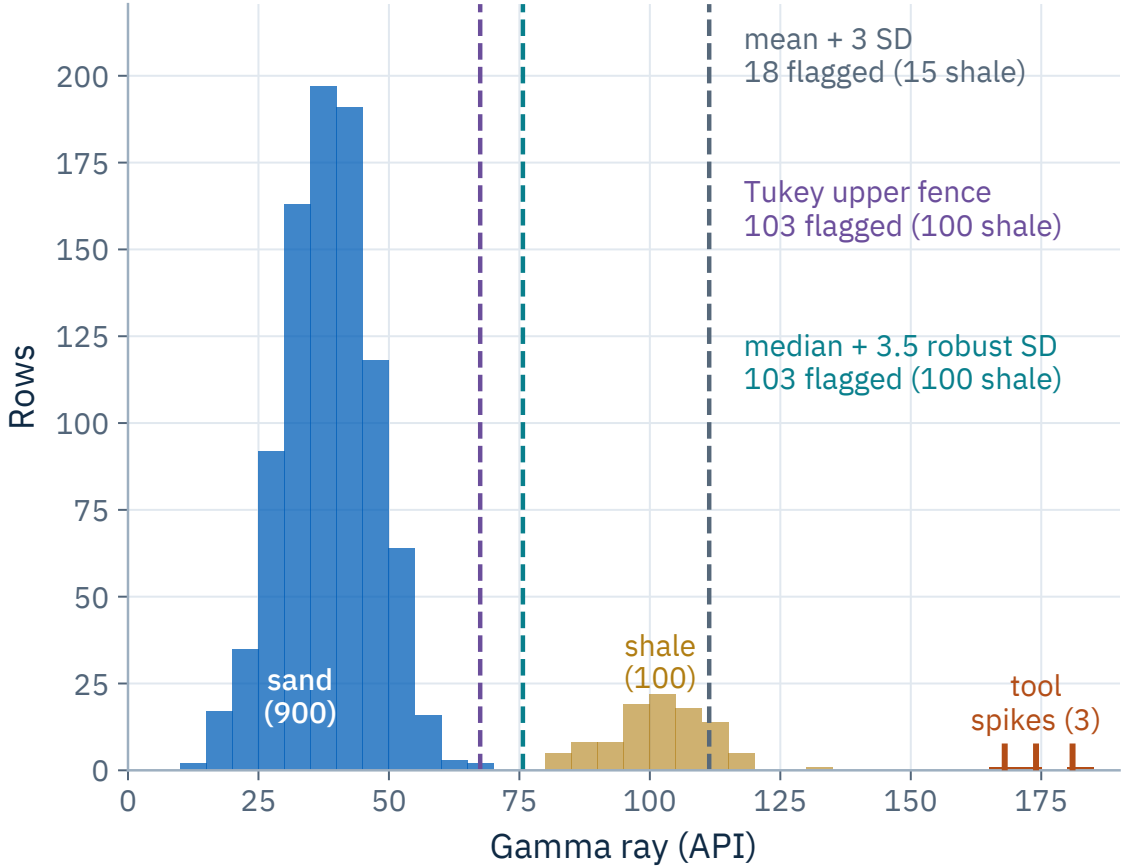
- 25 offset readings raise the SD from about 2 to 5.2 bar
- Classical |z| > 3 flags none of them: masking
- Median and MAD barely move; robust |z| > 3.5 flags all 25



Synthetic readings (seed 3): 200 ordinary N(50, 2) and 25 offset N(65, 0.8).

Univariate fences assume a single population

- Tukey fences: below Q1 – 1.5 IQR or above Q3 + 1.5 IQR
- A 10% shale population sits beyond both robust fences
- 100 of the 103 robust flags are valid shale, not tool error
- Set fences per facies or regime, or use multivariate methods

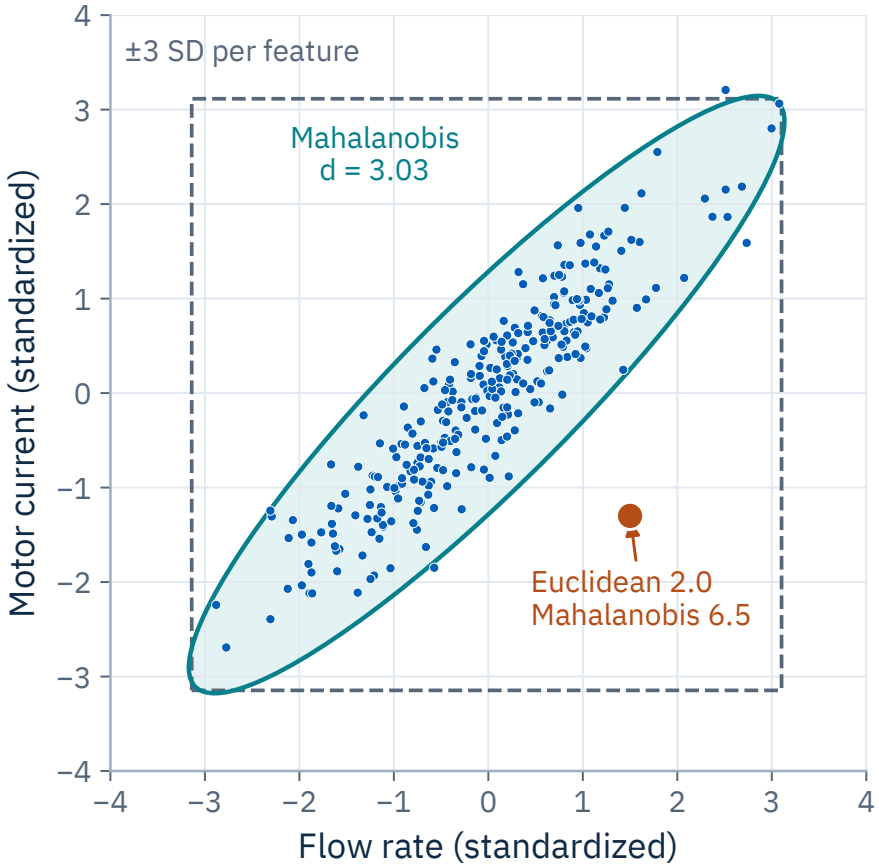


Synthetic GR-like mixture (seed 21): 900 sand rows N(38, 9), 100 shale rows N(102, 9), three tool spikes.

Joint outliers hide inside univariate limits

$$d_M(x) = \sqrt{(x-\mu)^T \Sigma^{-1} (x-\mu)}$$

- Each value of the orange point is within ±3 SD of its feature
- It breaks the flow–current relationship: $d_M = 6.5$
- For Gaussian data with two features, the 99% cut-off is 3.03
- Estimate μ and Σ robustly so outliers cannot mask themselves



Synthetic correlated features (seed 4): correlation 0.9, 300 rows; covariance estimated from the sample.

Physical rules catch what statistics cannot

Range

- Values beyond physical or tool limits
- e.g. negative flow, NPHI above 1

Rate of change

- Jumps faster than the process allows
- Spikes and dropouts in one curve

Flatline

- Stuck sensors repeat one value
- Rolling variance near zero

Consistency

- Related channels must agree
- e.g. caliper vs bit size, flow balance

◆ Rules encode engineering knowledge; keep them separate from learned scores.

PRACTICE

5 min · Individual

Compute a robust z-score by hand

Seven intake-pressure readings (bar): 48, 49, 50, 50, 51, 52, 71.

1. The mean is 53.0 and the SD is 8.04. Is 71 beyond $|z| = 3$?
2. Compute the median, the MAD, and the robust z of 71.
3. Can any reading in a sample of seven exceed $|z| = 3$?

PART 2 OF 7

02

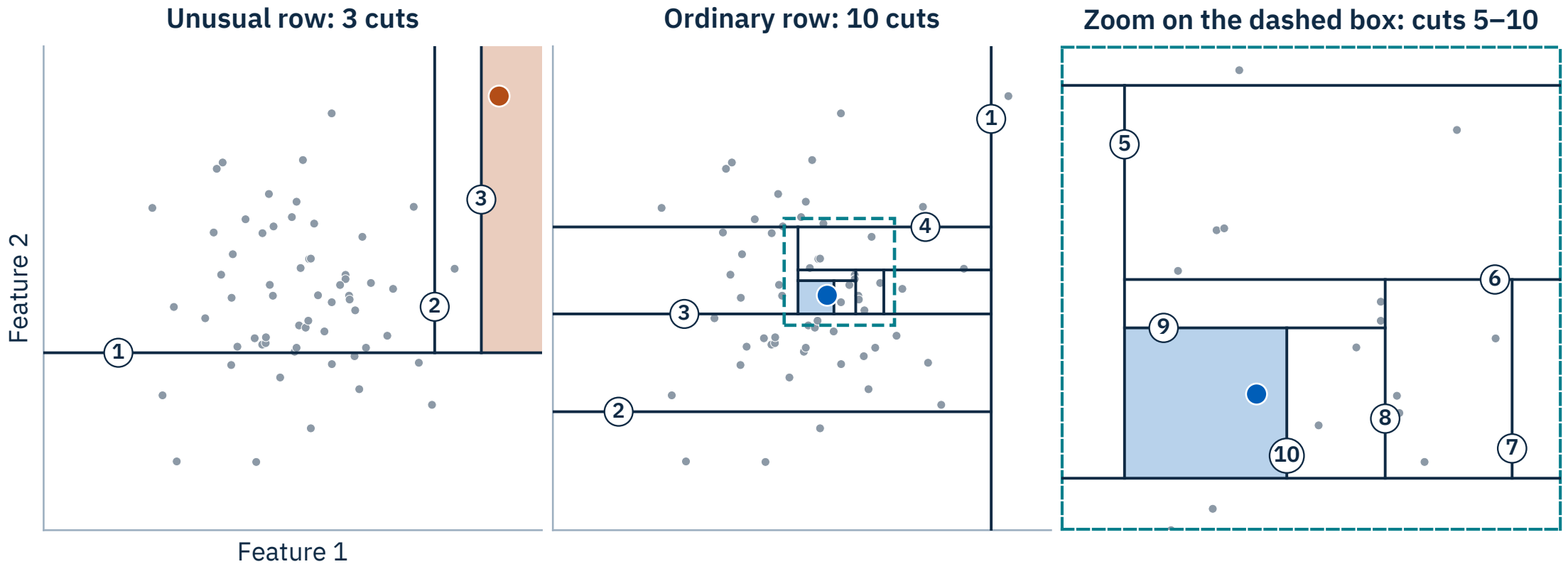
Detectors that learn normal behaviour

Each detector encodes a different idea of "far from normal".

- Isolation: how quickly random cuts separate a row
- Local density: LOF and the neighbour count
- Boundaries: One-Class SVM
- Reconstruction and forecast residuals for correlated and time-series data

Random cuts isolate unusual points in few steps

Each cut: a random feature, and a random value inside the current cell's range.

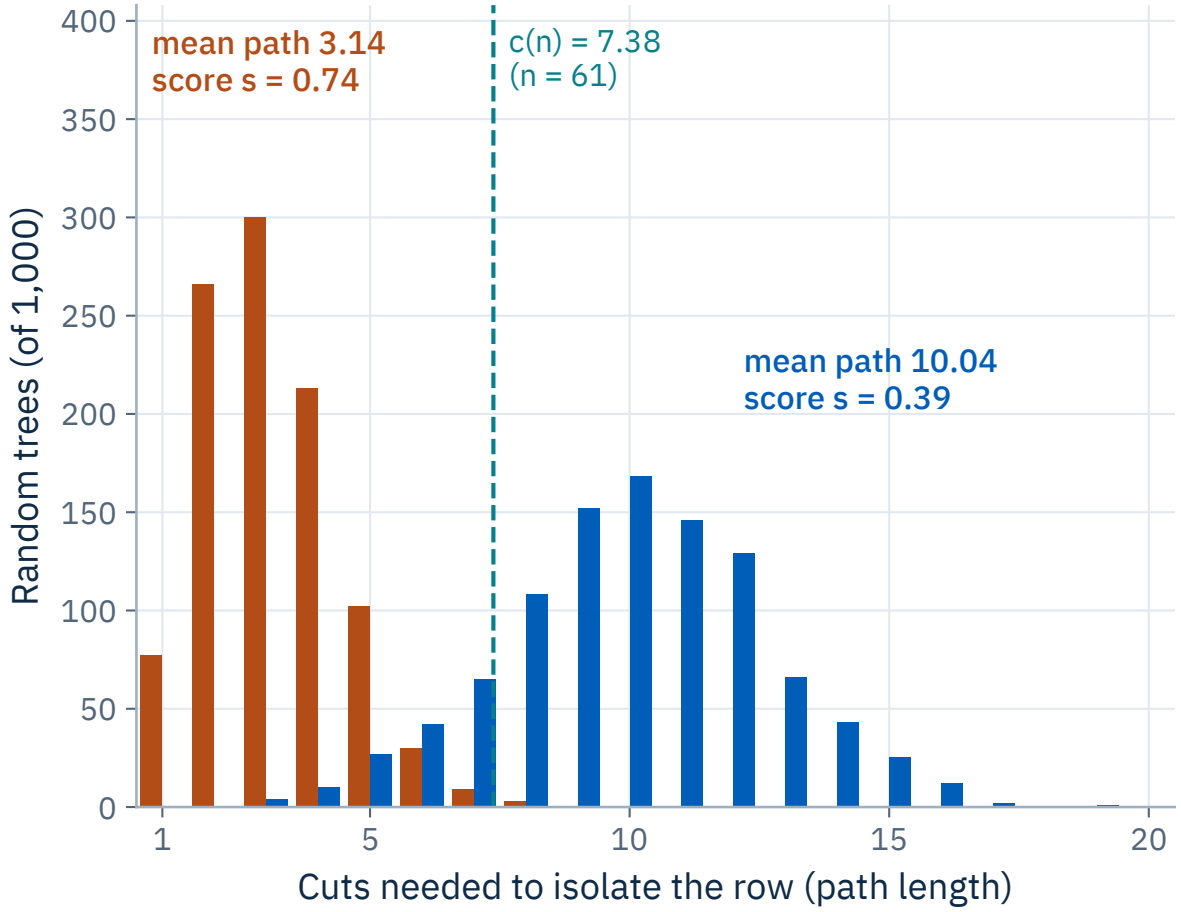


Synthetic 2-D sample (seed 2): 60 ordinary rows and one unusual row; one random isolation path for each highlighted row.

Isolation Forest turns short paths into high scores

$$s(x, n) = 2^{-E[h(x)]/c(n)}$$

- Average the path length $h(x)$ over many random trees
- $c(n)$ normalizes by the average path for n rows
- s near 1: isolated quickly; well below 0.5: ordinary
- Investigate ranked rows; validate any alert threshold

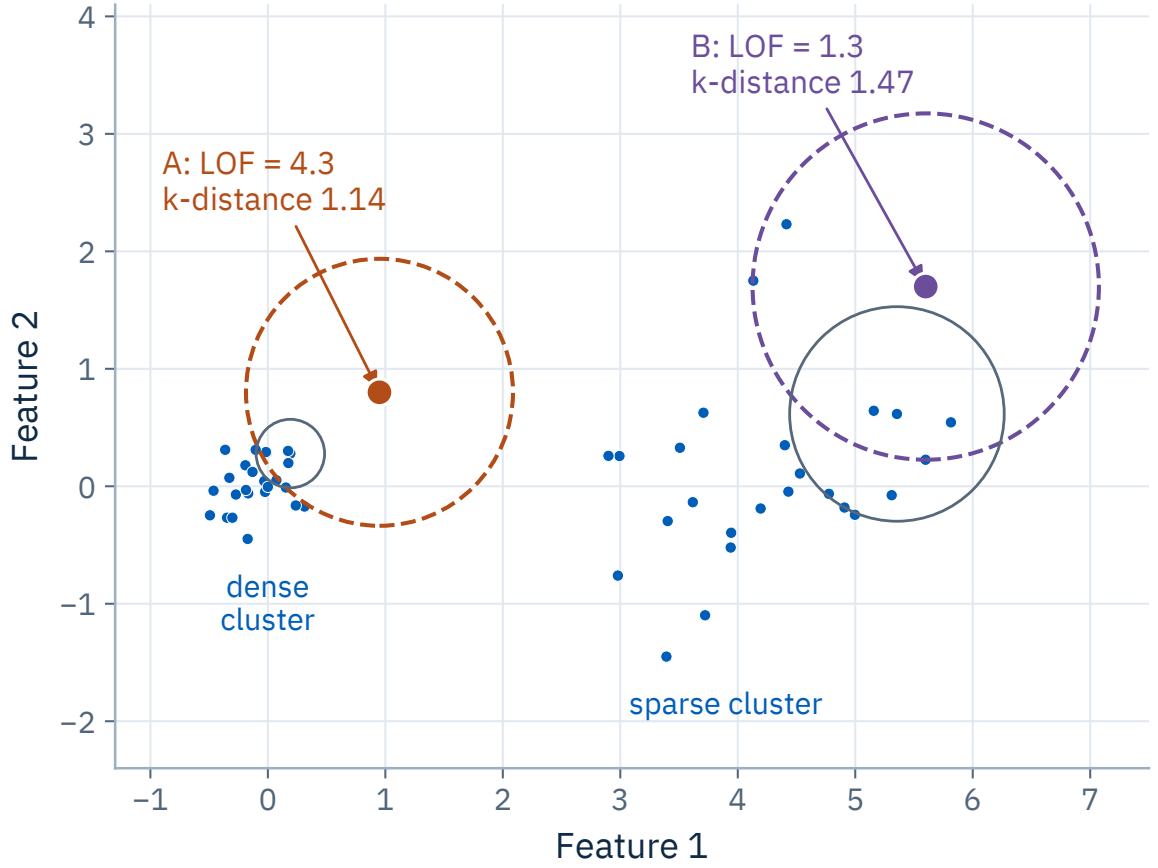


c(61) = 7.38 for the 61-row sample; 1,000 random paths per highlighted row.

Local Outlier Factor compares neighbourhoods

$$\mathrm{LOF}_k(p) = \frac{1}{k} \sum_{o \in N_k(p)} \frac{\mathrm{lrd}_k(o)}{\mathrm{lrd}_k(p)}$$

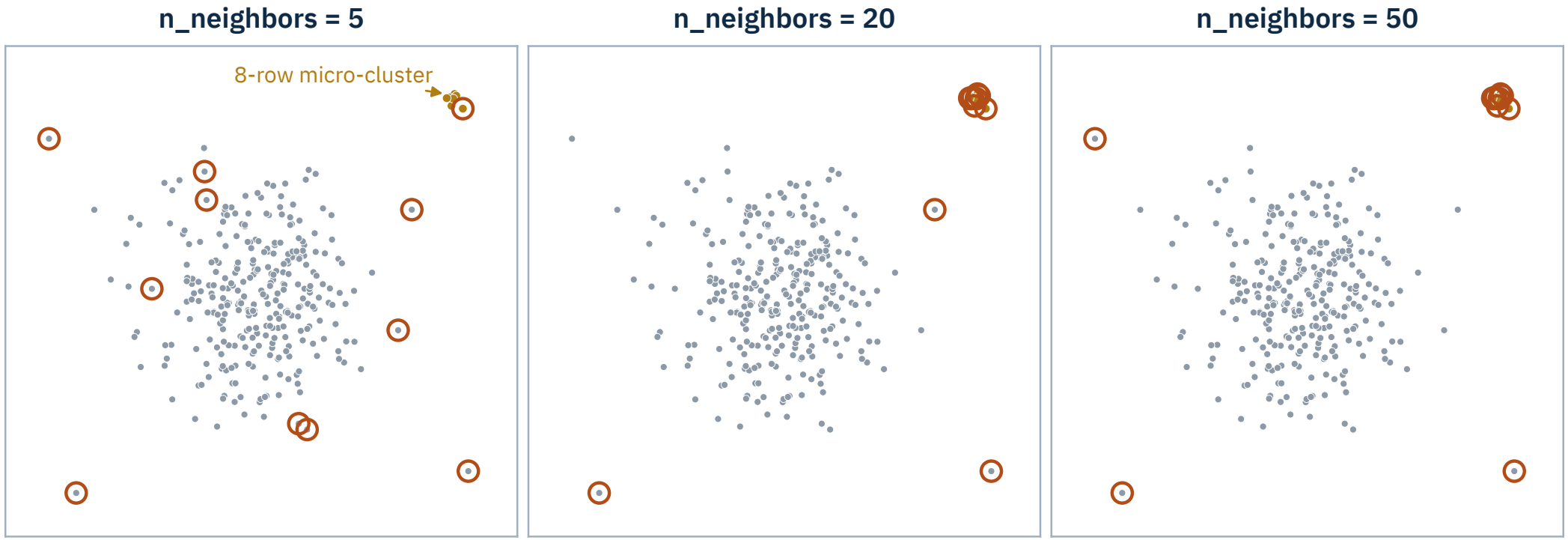
- lrd: inverse of the mean reachability distance to the k neighbours
- LOF near 1: as dense as its neighbours; well above 1: sparser
- A is closer to its neighbours than B, yet A is the local outlier
- Scaling and the neighbour count change the result



Synthetic clusters (seed 8); LocalOutlierFactor with n_neighbors = 6. Dashed circles: k-distances of A and B; grey circles: one neighbour of each.

The neighbour count decides what counts as local

Set n_neighbors above the size of any group of rows that should still count as unusual.

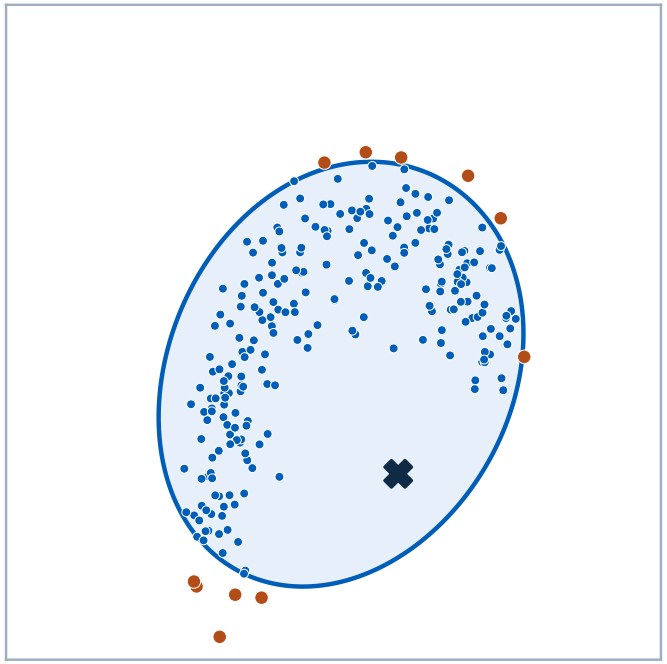


Synthetic data (seed 12): 300 ordinary rows, an 8-row micro-cluster, and 3 isolated rows; rings mark the 11 highest LOF scores.

One-Class SVM estimates a boundary

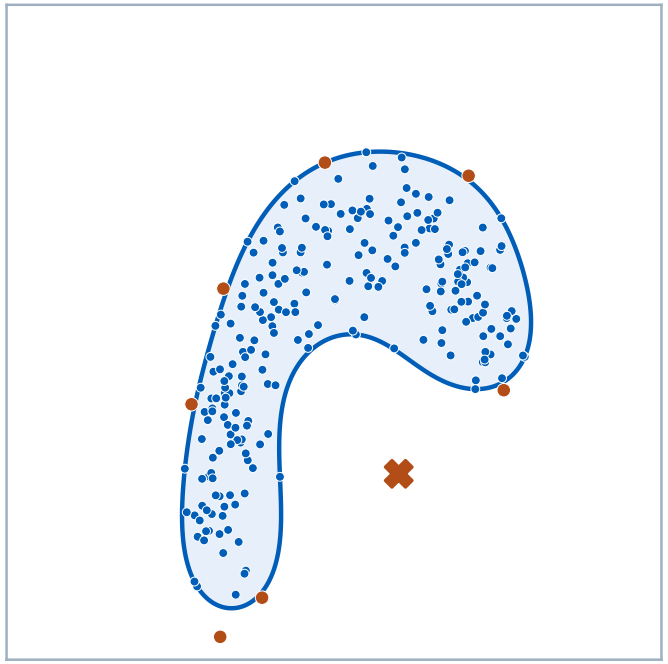
nu caps the share of training rows outside; gamma sets how tightly it wraps.

gamma = 0.05



new ordinary rows flagged: 7.9%
interior probe (X) flagged: no

gamma = 0.5



new ordinary rows flagged: 8.6%
interior probe (X) flagged: yes

gamma = 5

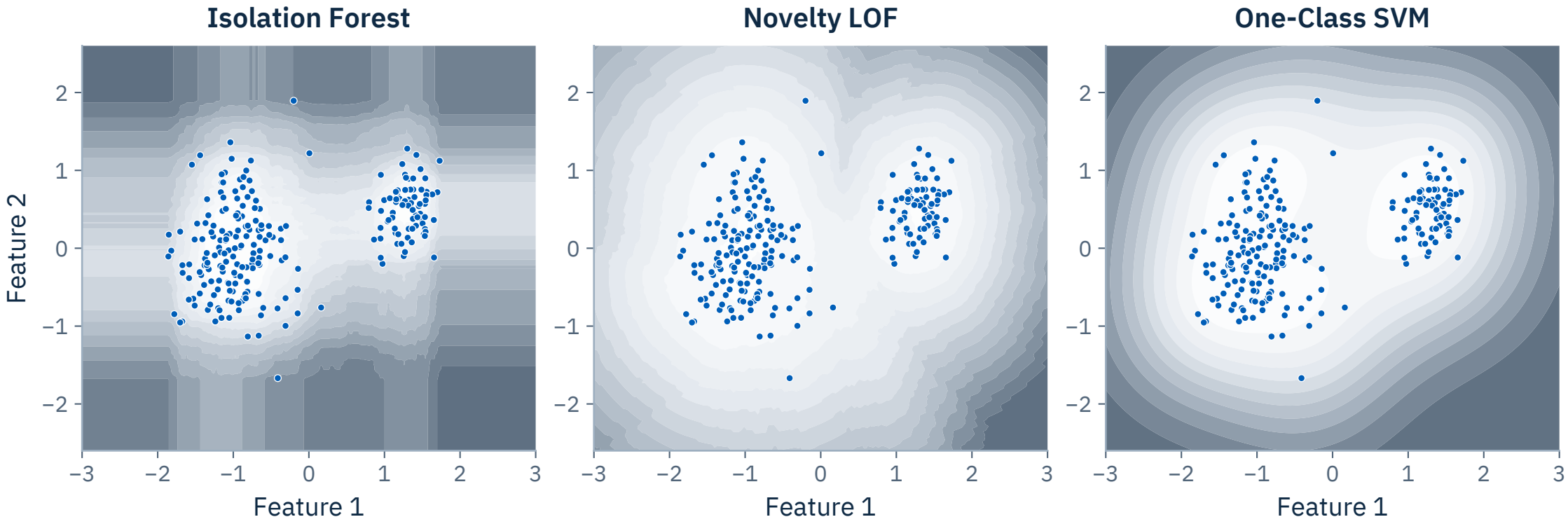


new ordinary rows flagged: 23.8%
interior probe (X) flagged: yes

Synthetic arc-shaped reference data (seed 6, standardized); RBF kernel, nu = 0.05. Flag rates measured on 2,000 new rows from the same generator.

Methods rank unusual observations differently

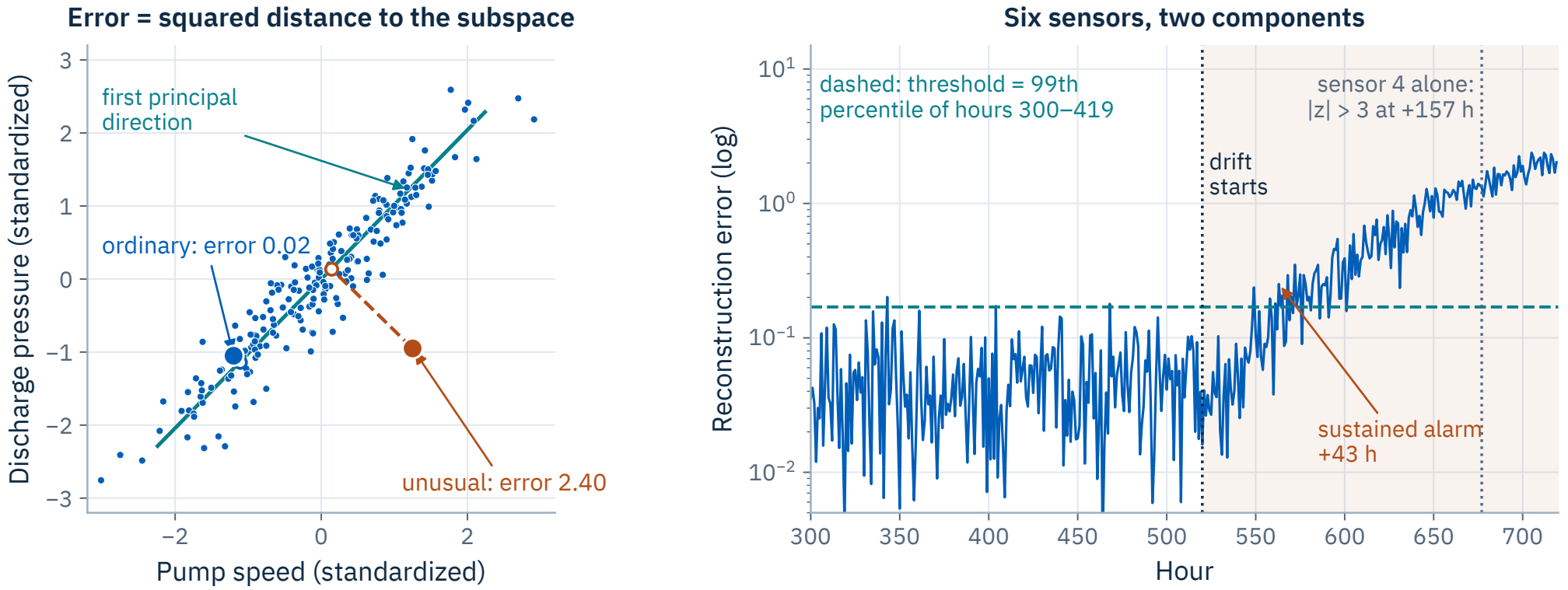
Darker background = more unusual under each model; blue points: the reference rows



Three detectors trained on the same synthetic reference set (seed 42). Backgrounds show model scores, not fault probabilities.

Reconstruction error flags broken relationships

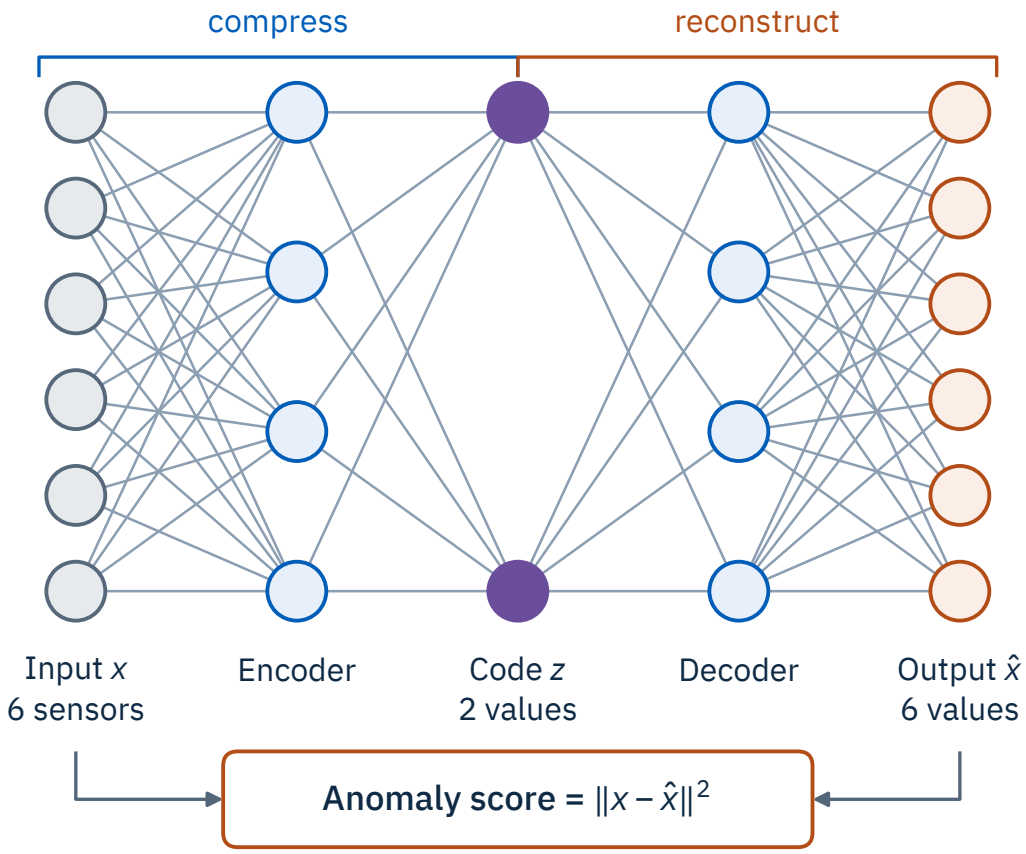
PCA (Module 4) keeps the correlations; rows that break them reconstruct badly.



Left: synthetic correlated pair (seed 13). Right: six synthetic channels (seed 32); PCA fitted on hours 0–299, threshold from hours 300–419, sensor 4 drifts from hour 520.

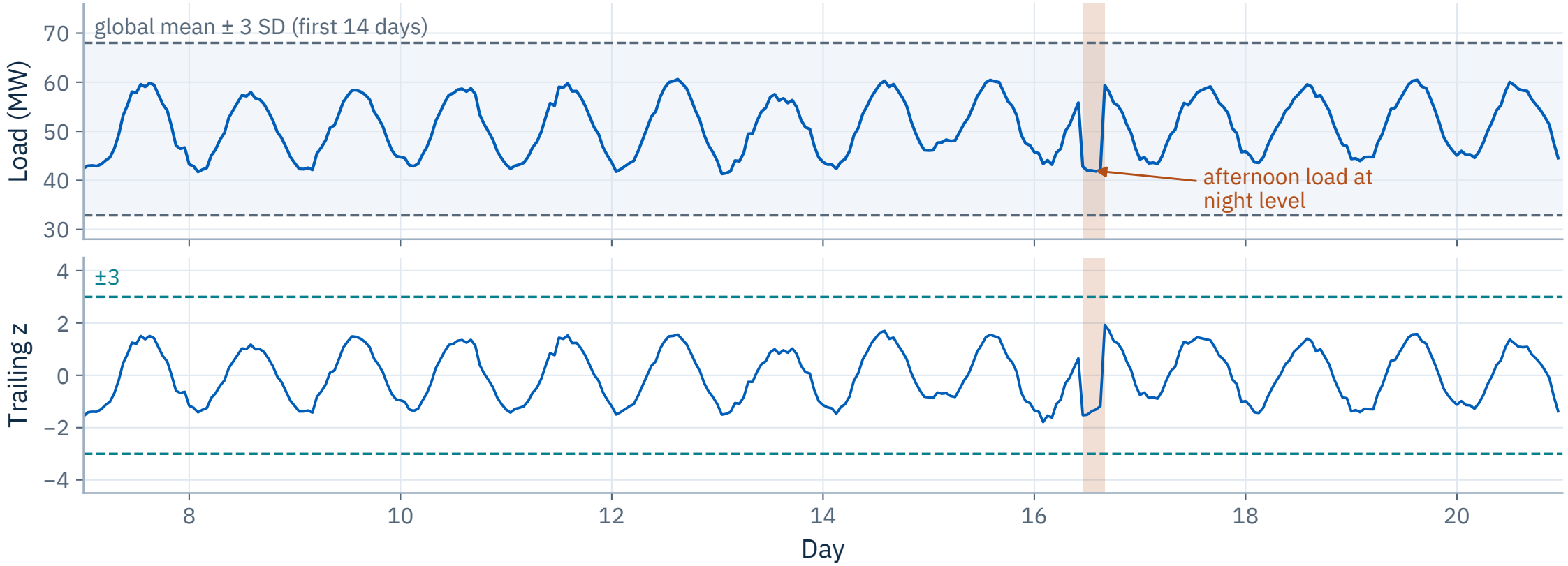
Autoencoders learn nonlinear reconstructions

- The encoder compresses; the decoder rebuilds the input
- Train on normal operation only; score by reconstruction error
- A linear autoencoder with squared error spans the PCA subspace
- Needs more data and tuning: see the neural network course



Schematic undercomplete autoencoder (6-4-2-4-6 units).

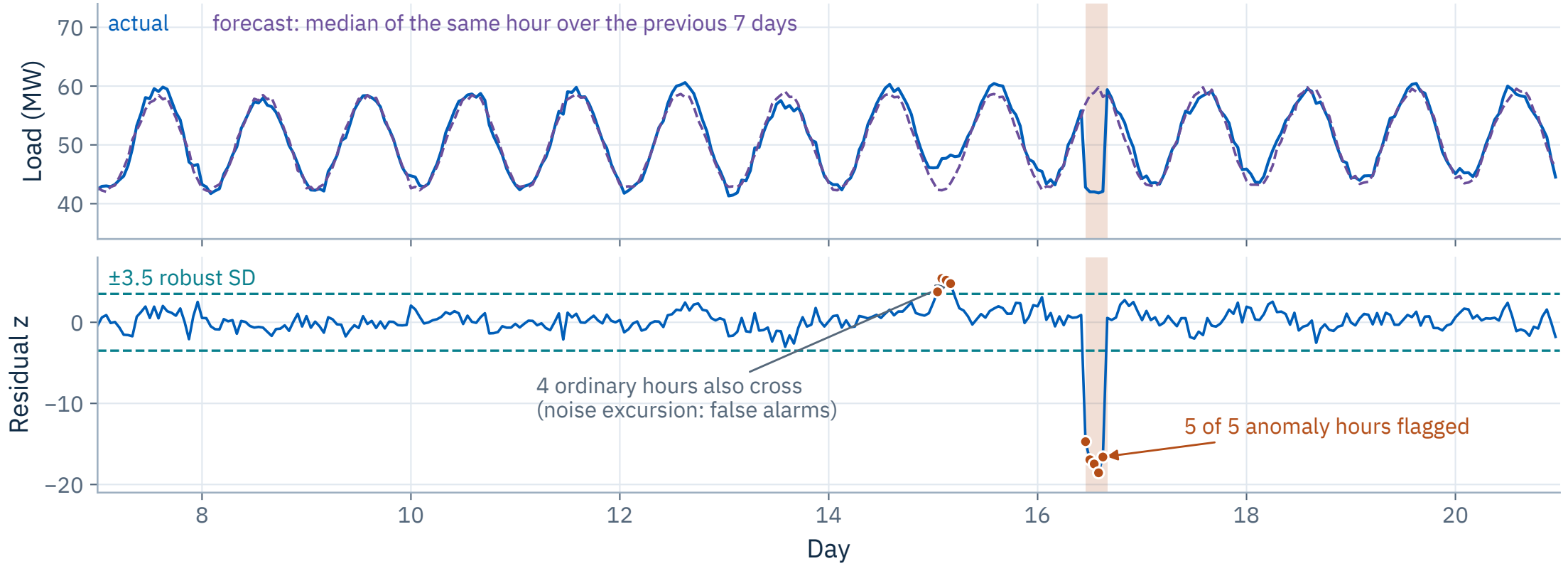
Global limits and rolling z-scores miss daily context



Synthetic hourly load (seed 17); five afternoon hours near day 16.5 set to night level. Top: mean ± 3 SD of the first 14 days. Bottom: z-score against the trailing 24 hours.

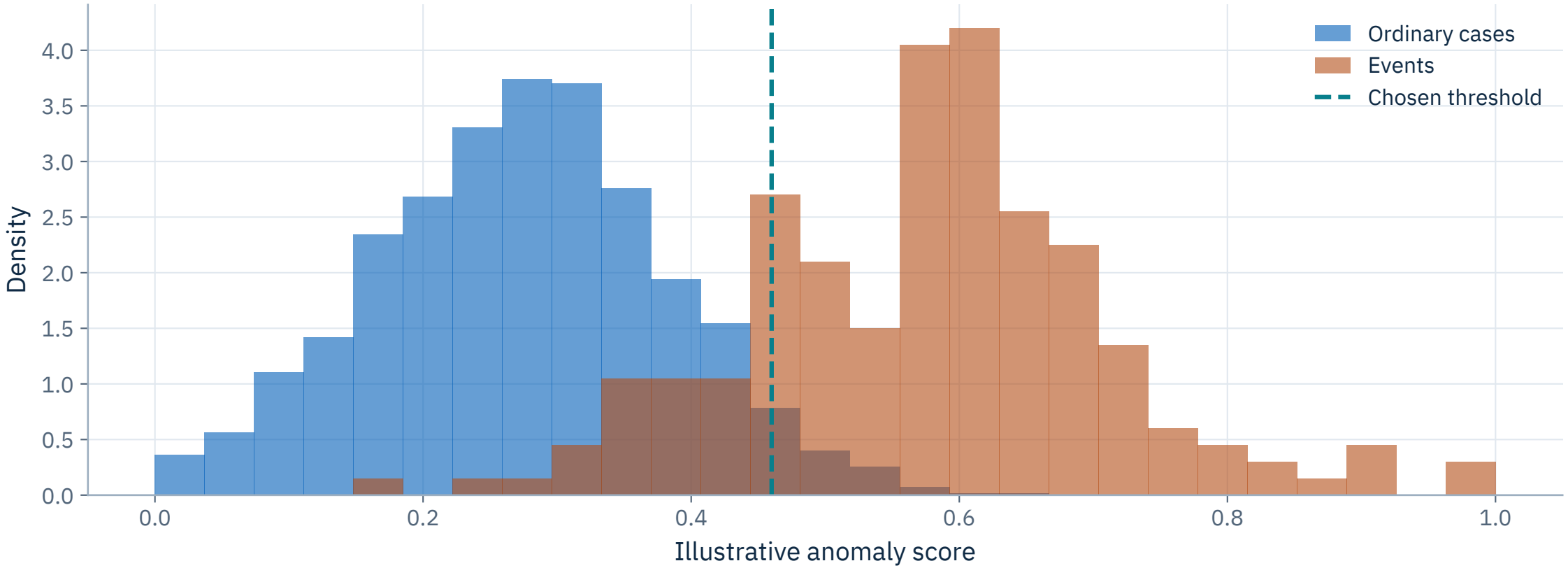
Forecast residuals expose contextual anomalies

Compare each hour with what that hour usually looks like, then threshold the residual.



Same series. Forecast: median of the same hour over the previous 7 days; residuals scaled by the median and MAD of days 7–14; threshold 3.5.

A score and an alert are separate outputs



Simulated score distributions (seed 42), not a fitted detector, with a chosen threshold. The overlap creates both missed events and false alarms.

Score signs differ across scikit-learn detectors

Estimator	Lower score_samples means	predict for outliers	Watch for
IsolationForest	More abnormal	−1	decision_function subtracts offset_
LocalOutlierFactor	More abnormal	−1	Only with novelty=True; else fit_predict
OneClassSVM	More abnormal	−1	Scale features; tune nu and gamma
EllipticEnvelope	More abnormal	−1	Assumes one elliptical cloud

Convert to "higher = more unusual" once, in one place, and test it.

Combine detectors by rank, not by raw score

```
from scipy.stats import rankdata

scores = {                          # higher = more unusual
    "iforest": -iso.score_samples(X),
    "lof": -lof.negative_outlier_factor_,
    "ocsvm": -ocsvm.decision_function(X),
}
pct = {k: rankdata(v) / len(v)   # 0-1 percentile ranks
       for k, v in scores.items()}
combined = np.mean(list(pct.values()), axis=0)
review = np.argsort(-combined)[:50]   # top 50 to review
```

- Raw scores differ in scale and sign across methods
- Percentile ranks put them on one 0–1 scale
- The average rank is a review order, not a probability
- Keep each method's rank to explain why a row was flagged

Fit the detector on a reference period only

```
import numpy as np
from sklearn.ensemble import IsolationForest
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler

ref = df[df["time"] < cutoff]      # reviewed, ordinary period
new = df[df["time"] >= cutoff]

det = make_pipeline(StandardScaler(),
                    IsolationForest(n_estimators=300, random_state=0))
det.fit(ref[features])

s_ref = -det.score_samples(ref[features])   # higher = more unusual
threshold = np.quantile(s_ref, 0.995)        # a start; validate it
alerts = -det.score_samples(new[features]) > threshold
```

PRACTICE

4 min · Pairs

Contamination is not a discovered prevalence

An analyst sets contamination = 0.02 and reports that 2% of the equipment is faulty.

1. What did the parameter actually control?
2. What evidence is missing from the conclusion?

Choose a first detector by its assumptions

Method	Assumes	Good first use	Watch for
Robust z / IQR	One population per feature	Single-sensor quality control	Mixtures, context
Mahalanobis	One elliptical cloud	Correlated channels	Masking: fit robustly
Isolation Forest	Anomalies few and different	Tabular baseline	Irrelevant features
LOF	Local density matters	Clusters of different density	n_neighbors, scaling
One-Class SVM	Compact normal region	Clean reference, modest size	gamma, nu, runtime
PCA reconstruction	Normal rows near a subspace	Many correlated sensors	Nonlinear relationships

LABORATORY

65 min

Lab 1: statistical rules and Isolation Forest

- Part 1: load, plot one well, drop incomplete rows; record the rows and wells lost
- Part 2: flag |z| > 3 and IQR outliers; compare the two sets
- Part 3: fit Isolation Forest (contamination 0.05) and plot its score track
- Sweep contamination from 0.01 to 0.10 and explain what changes

NOTEBOOK

`exercises/anomaly_detection/Anomaly_Detection_Student.ipynb`

EVIDENCE TO SUBMIT

- Row and well counts before and after cleaning
- z-score versus IQR overlap counts, with one reviewed interval
- One sentence on what contamination controls

DISCUSSION

Cleaning silently removed two of five wells

Reference run of the Lab 1 notebook (scikit-learn 1.7):

27,845
rows loaded

5,493
rows complete in the six curves

2 of 5
wells lost: DT missing in all rows

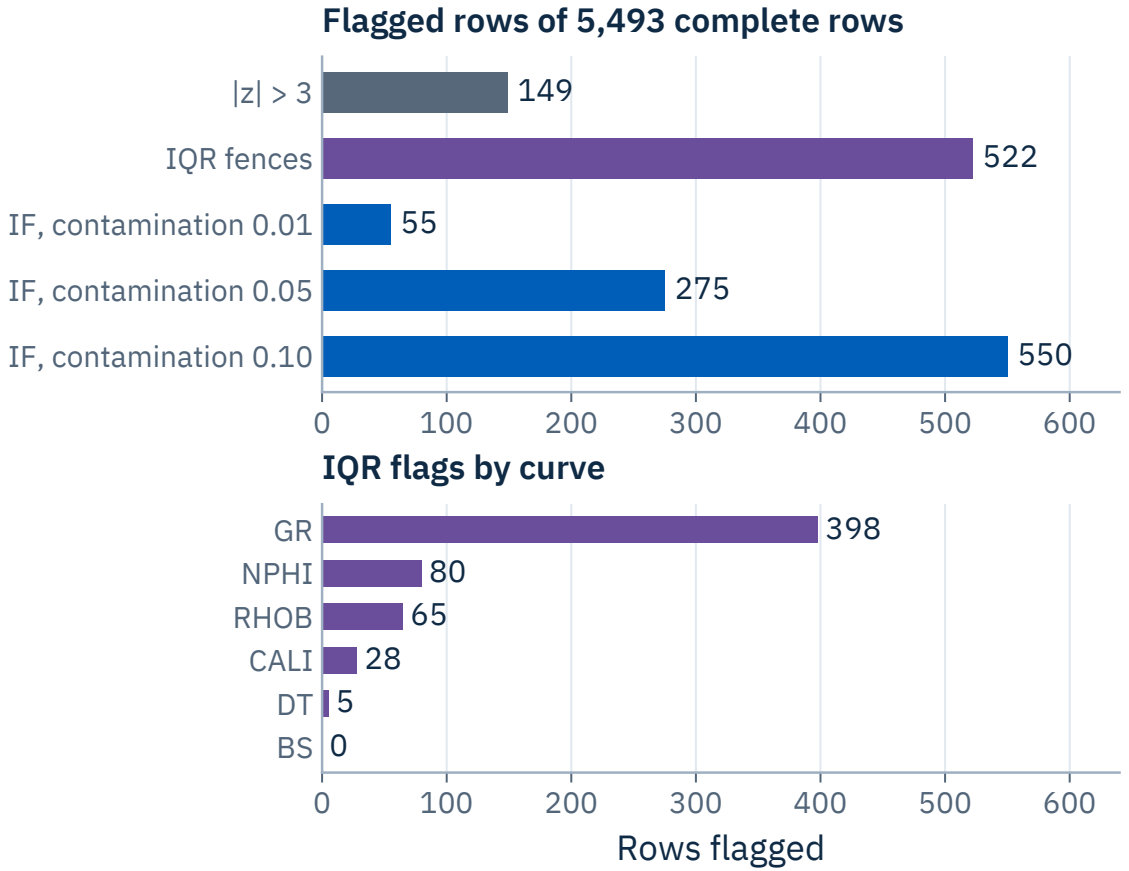
0
spread in BS: constant 8.5 in

- Record what cleaning removed before interpreting any flag.

DISCUSSION

Rules disagree, and contamination fixes the count

- Every |z| > 3 row is also an IQR row; IQR adds 373 more
- 398 IQR flags are high GR: shale-rich rows (median VSH 0.79)
- Isolation Forest flags exactly 1%, 5%, and 10% of the rows
- A higher contamination only lowers the threshold on one ranking



Computed from well_logs_data.csv (course data): 5,493 complete rows; IsolationForest(n_estimators=100, random_state=42).

PART 3 OF 7

03

Evaluating rare events

Useful alerts require denominators, thresholds, and event definitions.

- Base rates and the precision of alerts
- ROC versus precision-recall under imbalance
- Thresholds from costs and from review capacity
- Event-level metrics, lead time, and alert fatigue

PRACTICE

4 min | Individual

Prevalence changes the meaning of a false alarm rate

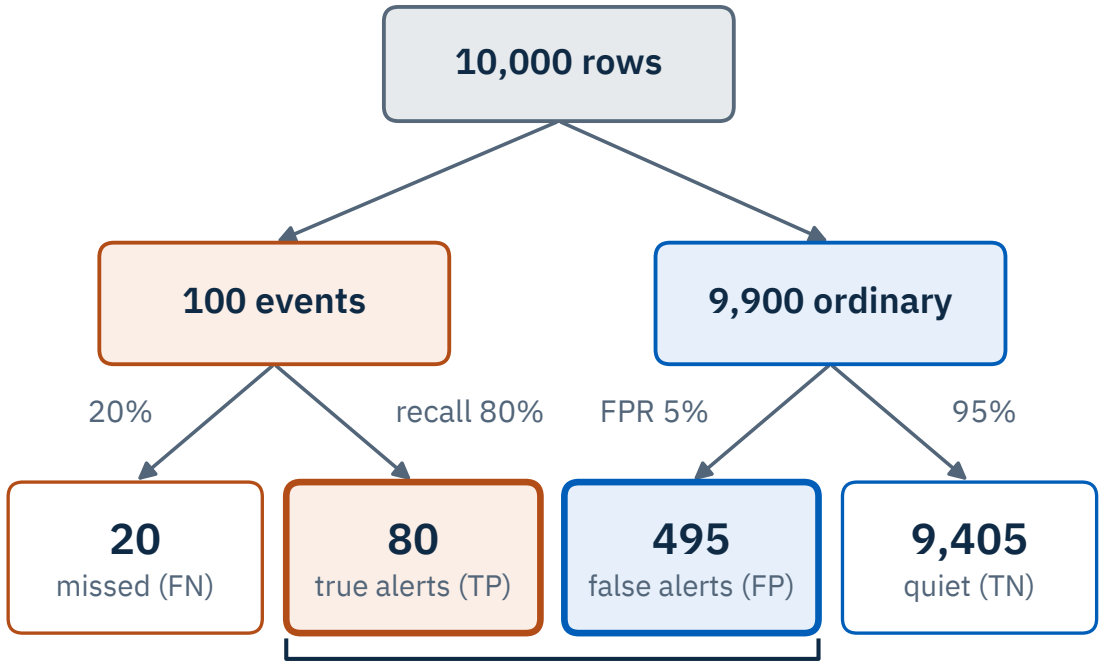
In 10,000 observations, 100 are events. A detector has 80% recall and a 5% false-positive rate.

1. How many true alerts and false alerts occur?
2. What fraction of alerts are true events?

DISCUSSION

Most alerts are false when events are rare

- 80 true alerts and 495 false alerts: precision 13.9%
- False alerts outnumber true ones about six to one
- The same detector at 10% prevalence: precision 64%
- Quote precision at the prevalence of deployment



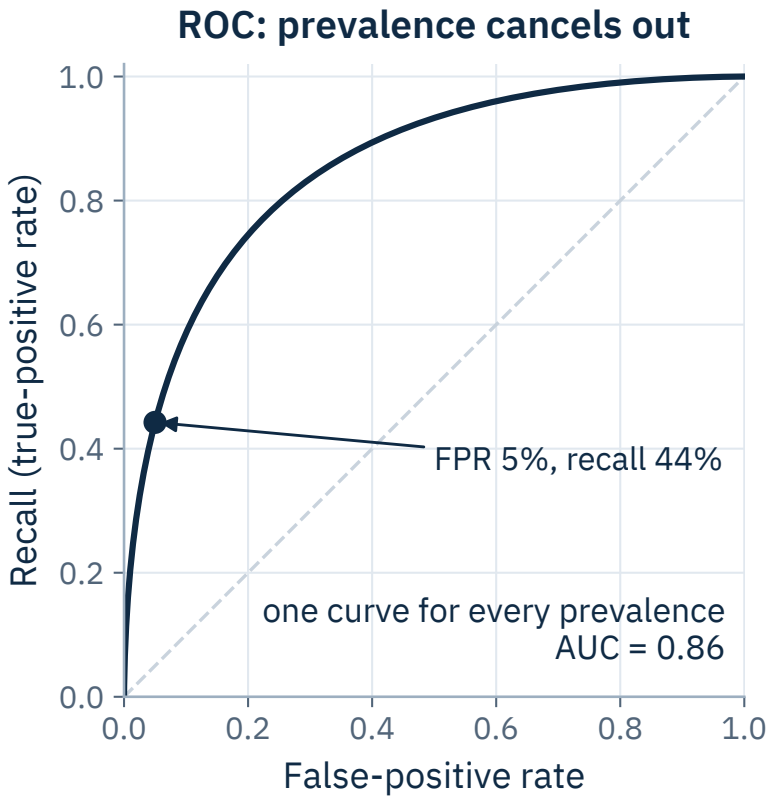
Alerts: 80 + 495 = 575

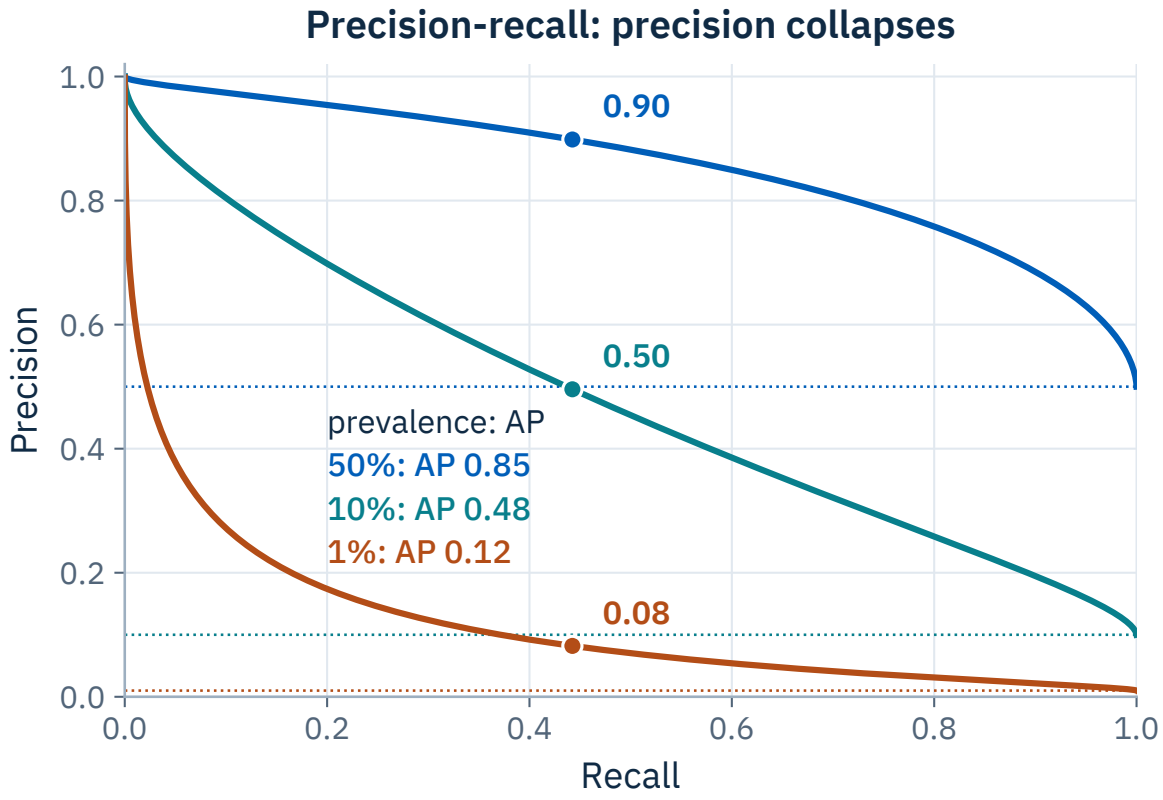
Precision = 80 / 575 = 13.9%

Exact arithmetic for 10,000 rows, 1% prevalence, recall 80%, false-positive rate 5%.

ROC curves hide what prevalence does to precision

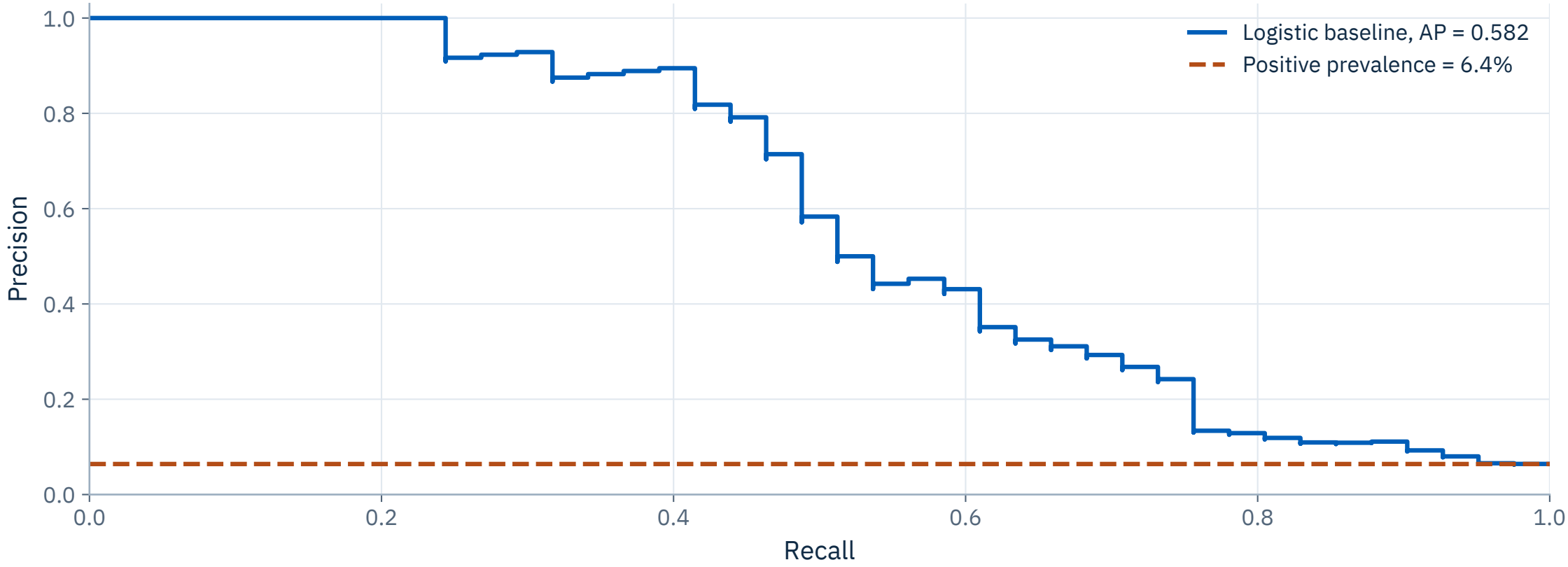
One detector, three prevalences: one ROC curve, three precision-recall curves.





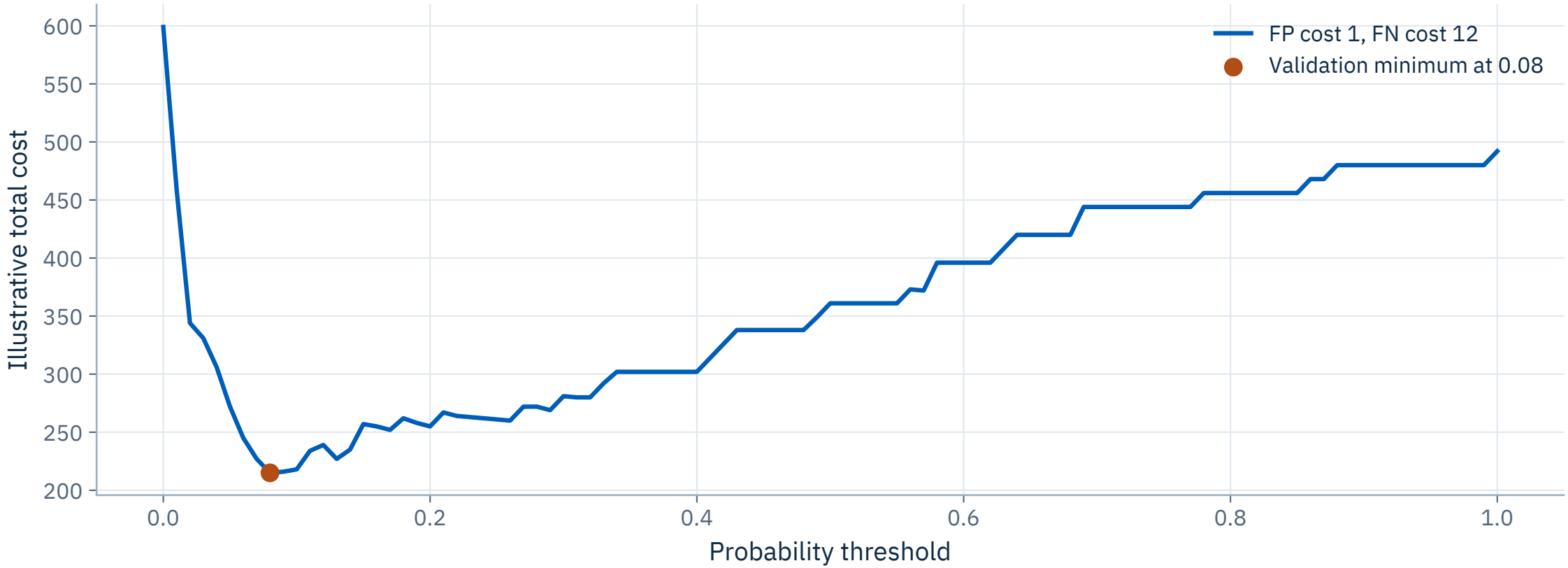
Exact binormal scores: ordinary N(0, 1), events N(1.5, 1). Dots mark the threshold with a 5% false-positive rate (recall 0.44) and the resulting precision.

Precision-recall exposes the alert trade-off



Logistic regression on a synthetic rare-event validation set (make_classification, seed 42). The dashed line is the observed positive prevalence; AP summarizes ranking across thresholds.

The threshold belongs to the decision



Cost computed on the same synthetic validation set (seed 42) with an illustrative cost ratio of 12 to 1. This is not an operational cost estimate.

With calibrated probabilities, costs set the threshold

$$t^* = \frac{C_{FP}}{C_{FP} + C_{FN}}$$

- Alert when the expected miss cost exceeds the false-alarm cost
- Costs 1 and 12 give $t^* = 1/13 \approx 0.077$
- Close to the validation minimum of 0.08 on the previous slide
- Valid only if the probabilities are calibrated: see Part 4

◆ Review capacity can override costs: set the threshold to an alert rate the team can act on.

Events differ from individual rows

Define the event

- Start, end, and grouping of each labelled event
- A warning window agreed before scoring

Score the event

- Event recall and useful warning time
- No credit for alerts after onset

Count false alerts per exposure

- Per asset-day or per pump-month
- Not per row

State the alert rules

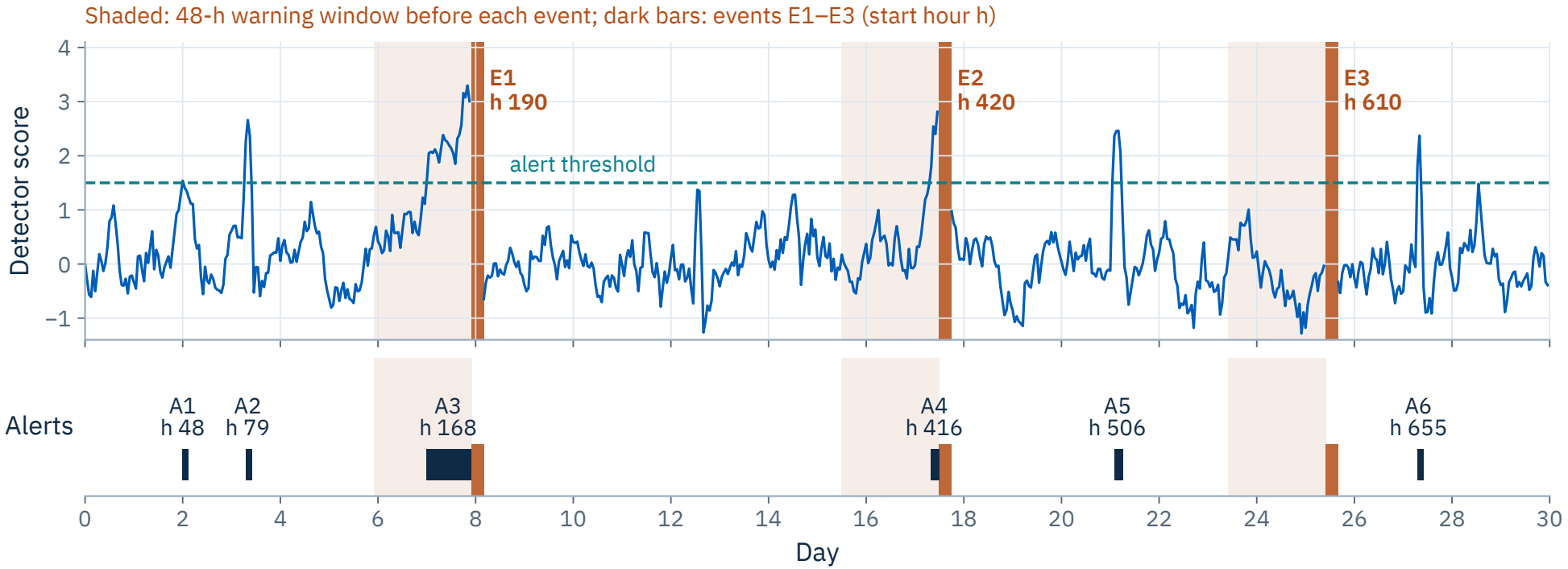
- Persistence, cooldown, deduplication
- Applied the same way in validation

PRACTICE

6 min · Pairs

Score an alert log at the event level

1. Which events were warned within 48 h, and with what lead time?
2. How many false alert episodes occurred per day?



Synthetic 30-day detector score (seed 22). Alert rows above the threshold are merged into episodes A1–A6 when gaps are 6 h or less; h = start hour.

DISCUSSION

Event metrics answer the operator's questions

Same alerts, counted by events instead of rows:

2 of 3

events warned within 48 h

22 h, 4 h

lead times for E1 and E2

0.13

false alert episodes per day

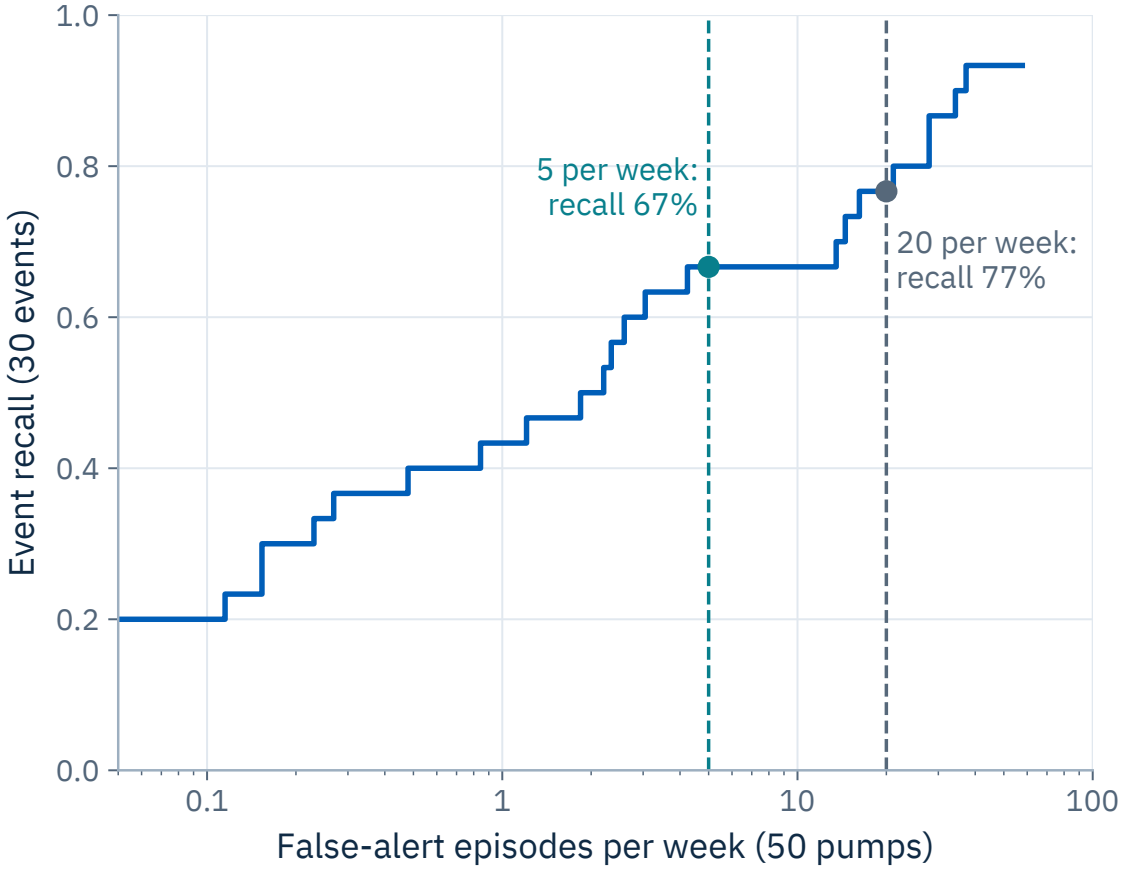
18%

row-level recall of the same alerts

- Row recall of 18% sounds poor; two of three events warned is the operational answer.

Alert capacity chooses the operating point

- Plot event recall against false alerts per week, not per row
- A team that reviews 5 false alerts a week gets 67% recall here
- Quadrupling the load to 20 a week buys only 77%
- Unreviewed alerts teach people to ignore the system



Synthetic fleet (seed 41): 50 pumps over 365 days, 30 events with precursors of random strength; alert episodes are runs of alert days outside warning windows.

Without labels, judge a detector by stratified review

Score band	Rows	Confirmed in a sample of 30	Estimated problems
Top: flagged	275	12	110
Middle	1,000	3	100
Bottom	4,218	1	141

Hypothetical review counts. Precision in the flagged band is 12 of 30 (40%), yet it holds only 110 of about 351 estimated problems (31%).

- Sample every band, unflagged rows included; review blind; weight each band by its size.

PART 4 OF 7

04

Imbalance, calibration, uncertainty

Rebalancing changes probabilities; few events limit claims.

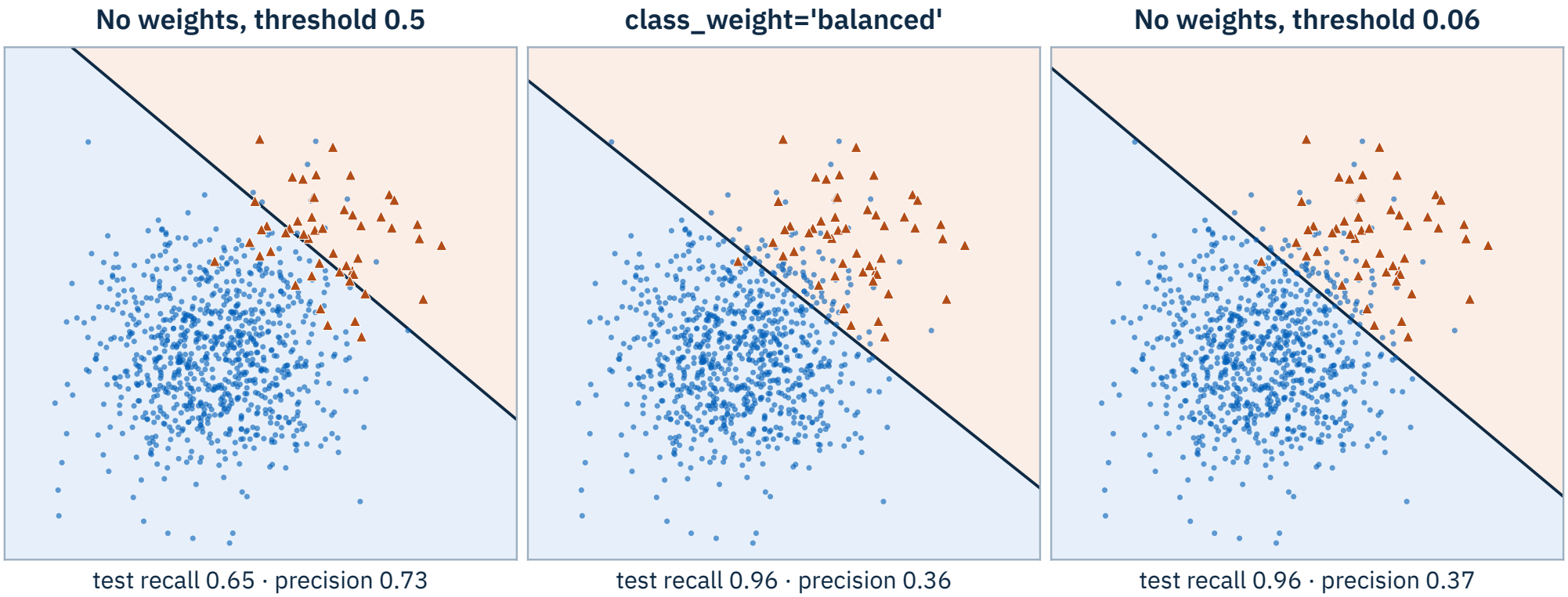
- Class weights, resampling, and SMOTE inside folds
- Calibration: reliability diagrams and recalibration
- Prior correction after undersampling
- Honest intervals with ten events

Each rebalancing strategy trades one risk for another

Strategy	Changes	Watch for
Class-weighted loss	Relative training penalty	Probability calibration changes
Random undersampling	Majority examples discarded	Loss of useful variation
Random oversampling	Minority examples repeated	Overfitting repeated cases
SMOTE	Interpolated minority features	Implausible combinations

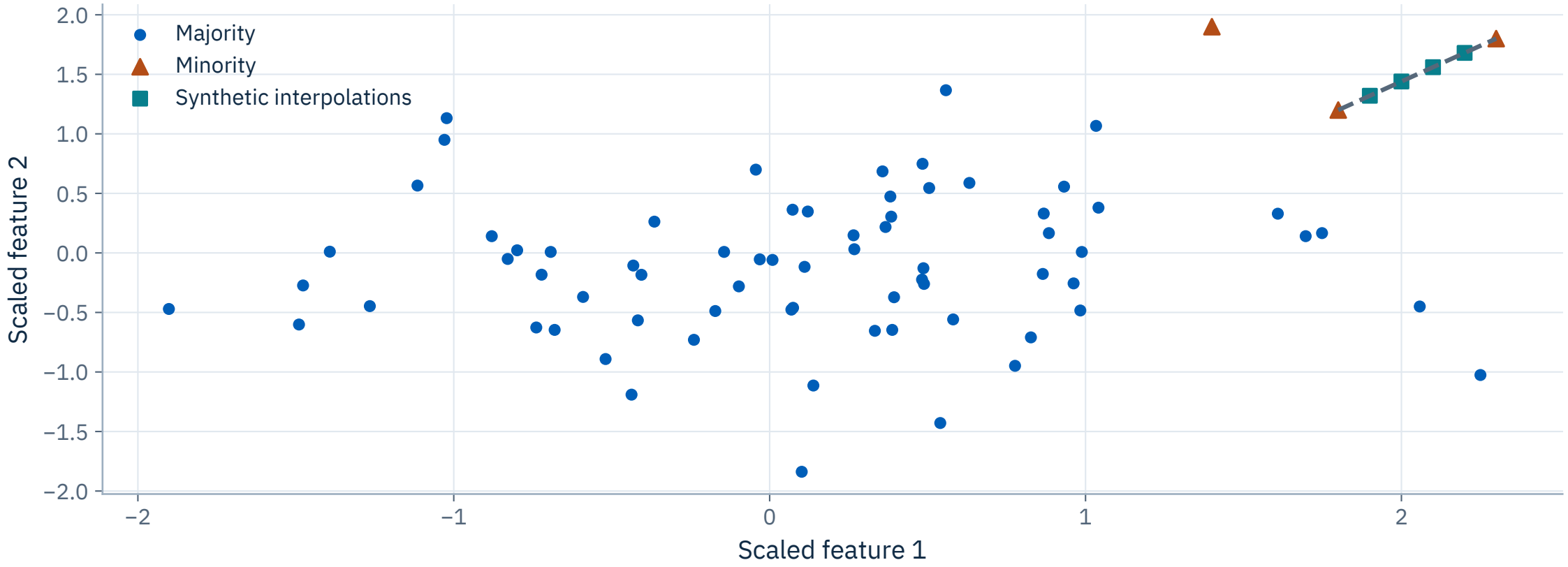
Class weights shift the decision boundary

Balanced weights move the boundary much as a lower threshold would.



Synthetic 2-D data (seed 9), 5% events, 50/50 stratified split; logistic regression. Right: unweighted model with the threshold that matches the balanced model's training recall.

SMOTE interpolates in a chosen feature space



Synthetic points (seed 42): one SMOTE-style interpolation between two minority observations. The segment is mathematical interpolation, not evidence of physical feasibility.

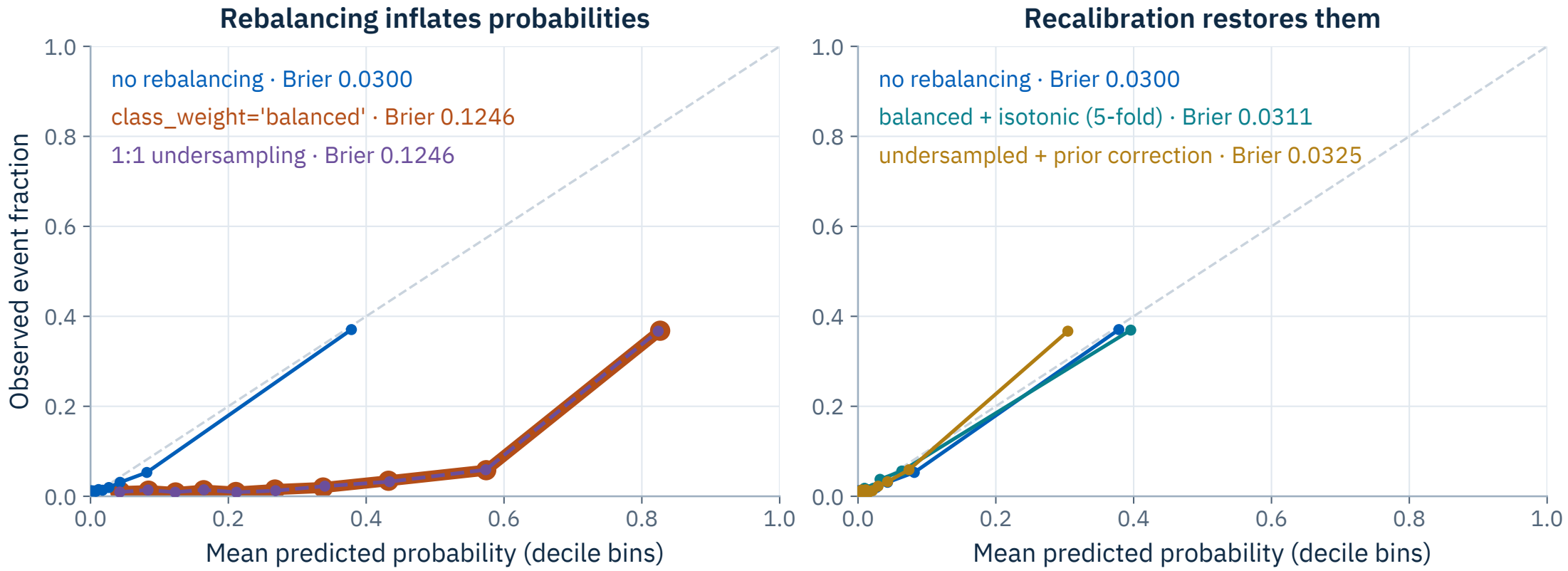
Resampling must happen inside training folds

```python
from imblearn.pipeline import Pipeline
from imblearn.over_sampling import SMOTE
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

model = Pipeline([
    ("scale", StandardScaler()),
    ("sample", SMOTE(random_state=42)),
    ("model", LogisticRegression(max_iter=1000))
])
```

Reweighting and undersampling distort probabilities

Mean probability 0.31 against an event rate of 0.055; the ranking barely moves.



Synthetic data (make_classification, seed 5, 40,000 rows, 5.5% events); logistic regression; reliability curves on the test half in decile bins.

Recalibrate on rows the model did not train on

```python
from sklearn.calibration import CalibratedClassifierCV
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import brier_score_loss
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler

lr = LogisticRegression(class_weight="balanced",
                        max_iter=1000)
base = make_pipeline(StandardScaler(), lr)
cal = CalibratedClassifierCV(base, method="isotonic",
                             cv=5)
cal.fit(X_train, y_train)

p = cal.predict_proba(X_test)[:, 1]
print(brier_score_loss(y_test, p))
```

- Each fold fits the model, then a calibrator on held-out rows
- method="sigmoid" is Platt scaling: needs fewer events
- Isotonic is more flexible but needs more events
- Check a reliability plot and the Brier score on test rows

Undersampling needs a prior correction

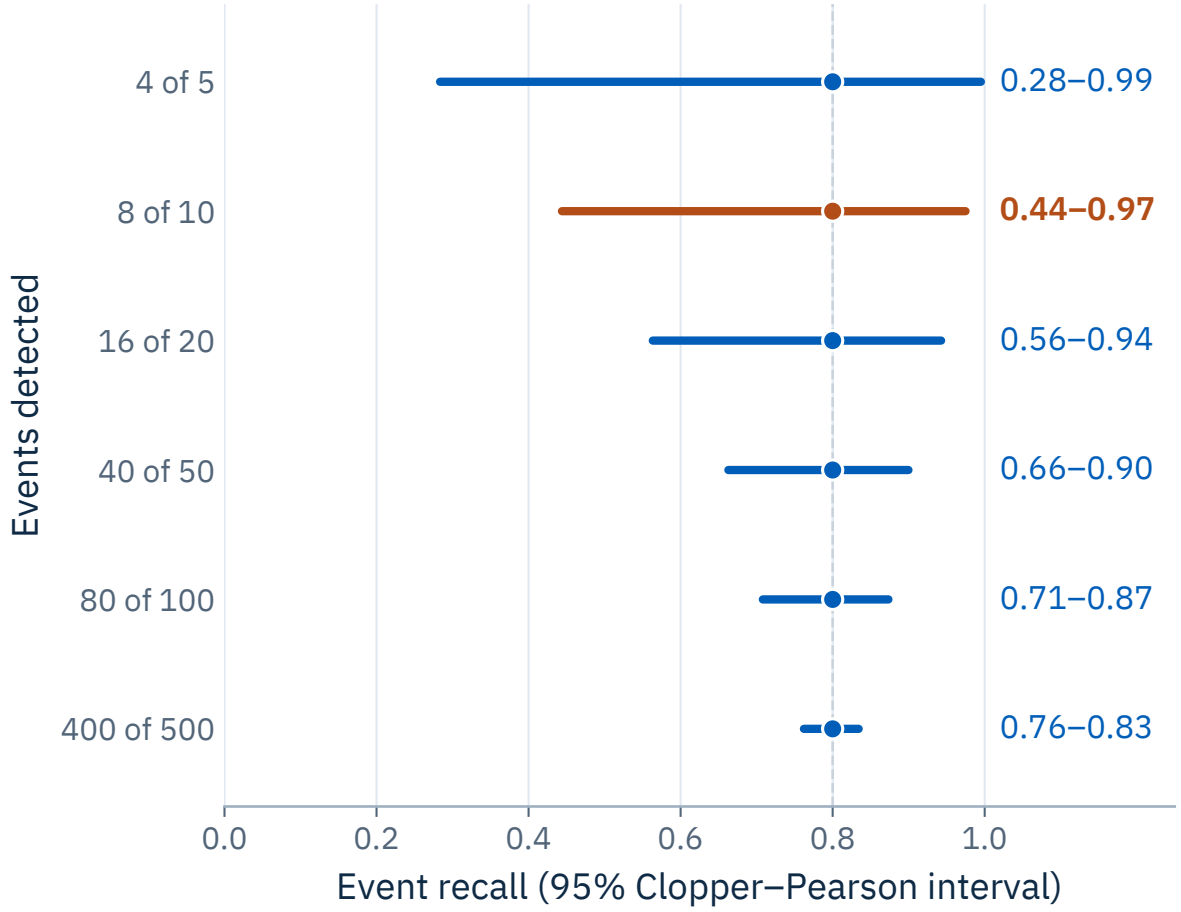
$$p = \frac{\beta p_s}{\beta p_s - p_s + 1}$$

- p_s: probability from the model trained on undersampled data
- β: fraction of majority rows kept (0.058 here)
- Restores the base rate: Brier 0.1246 falls to 0.0325
- Assumes the kept majority rows were sampled at random

◆ For class-weighted models there is no such shortcut: recalibrate on held-out rows.

Few labelled events require honest uncertainty

- Count independent events, not just labelled rows
- Review a sample of alerts and of non-alerts
- Resample whole events or assets, not rows
- State what the labels cannot estimate



Exact Clopper–Pearson 95% intervals for an observed recall of 80%.

PRACTICE

4 min · Pairs

Eight of ten events detected: what can you claim?

A detector warned of 8 of the 10 pump trips in the validation year.

1. Is "recall = 80%" a fair summary for the report?
2. What is the 95% interval, and what would 80 of 100 give?
3. Name one way the ten trips might not be independent.

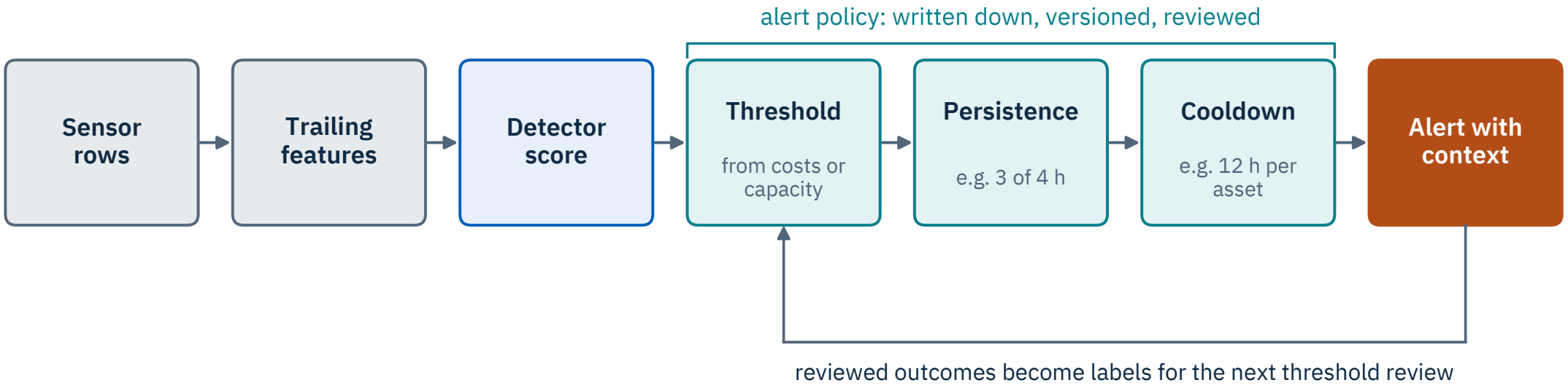
KEY TAKEAWAYS

Rare-event evidence checklist

- ✓ Keep the deployment prevalence in validation and test sets
- ✓ Report precision-recall and event-level metrics, not accuracy
- ✓ Freeze the threshold before the test; justify it by cost or capacity
- ✓ Check calibration after any weighting or resampling
- ✓ Quote intervals based on the number of independent events

An alert policy turns scores into accountable actions

Lab 2 ends with a written policy: each box in this chain needs a stated choice and its evidence.



Schematic. Persistence and cooldown values are illustrative choices, not recommendations.

LABORATORY

75 min

Lab 2: compare detectors and write an alert policy

- [ ] Part 4: LOF with n_neighbors from 5 to 100; compare flagged sets, not counts
- [ ] Part 5: One-Class SVM (nu = 0.05); plot its score track
- [ ] Part 6: overlap matrix, votes, and a rank-averaged review list
- [ ] Write a half-page alert policy with a threshold rationale

NOTEBOOK

`exercises/anomaly_detection/Anomaly_Detection_Student.ipynb`

EVIDENCE TO SUBMIT

- Overlap matrix of the five methods
- Jaccard overlap of LOF sets across n_neighbors
- Alert policy: score, threshold, rules, capacity, limits

Detectors agree on a core and differ at the margins

- IF and One-Class SVM share 173 rows; LOF overlaps least
- LOF always flags 275 rows, but which rows changes with n_neighbors
- 24 rows are flagged by all five methods; 322 by two or more
- Agreement is not validation: no label confirms any flag yet

	|z| > 3	IQR	IF	LOF	OCSVM
|z| > 3	149	149	105	26	84
IQR	149	522	164	48	132
IF	105	164	275	68	173
LOF	26	48	68	275	74
OCSVM	84	132	173	74	279

diagonal: rows flagged by each method;
off-diagonal: rows flagged by both

Computed from well_logs_data.csv (course data), 5,493 complete rows, lab settings; scikit-learn 1.7.

DISCUSSION

A defensible alert policy shows its evidence

Same detector, same data: only the second policy can be checked and improved.

✗ Weak policy

- "Flag the top 5% of Isolation Forest scores"
- No unit, no capacity, no review loop
- Reports that 5% of the depths are bad

vs

✓ Defensible policy

- Score, threshold, and why: cost or capacity
- Persistence, cooldown, routing, and owner
- Evidence, known limits, and a review date

◆ A policy is defensible when a colleague could check every number in it.

PART 5 OF 7

05

Honest and efficient model selection

Every choice made on validation data is part of the model.

- Tuning plans, grid search, and random search
- Successive halving and Bayesian optimization
- Selection bias and nested cross-validation
- Comparing models on paired folds; the one-SE rule

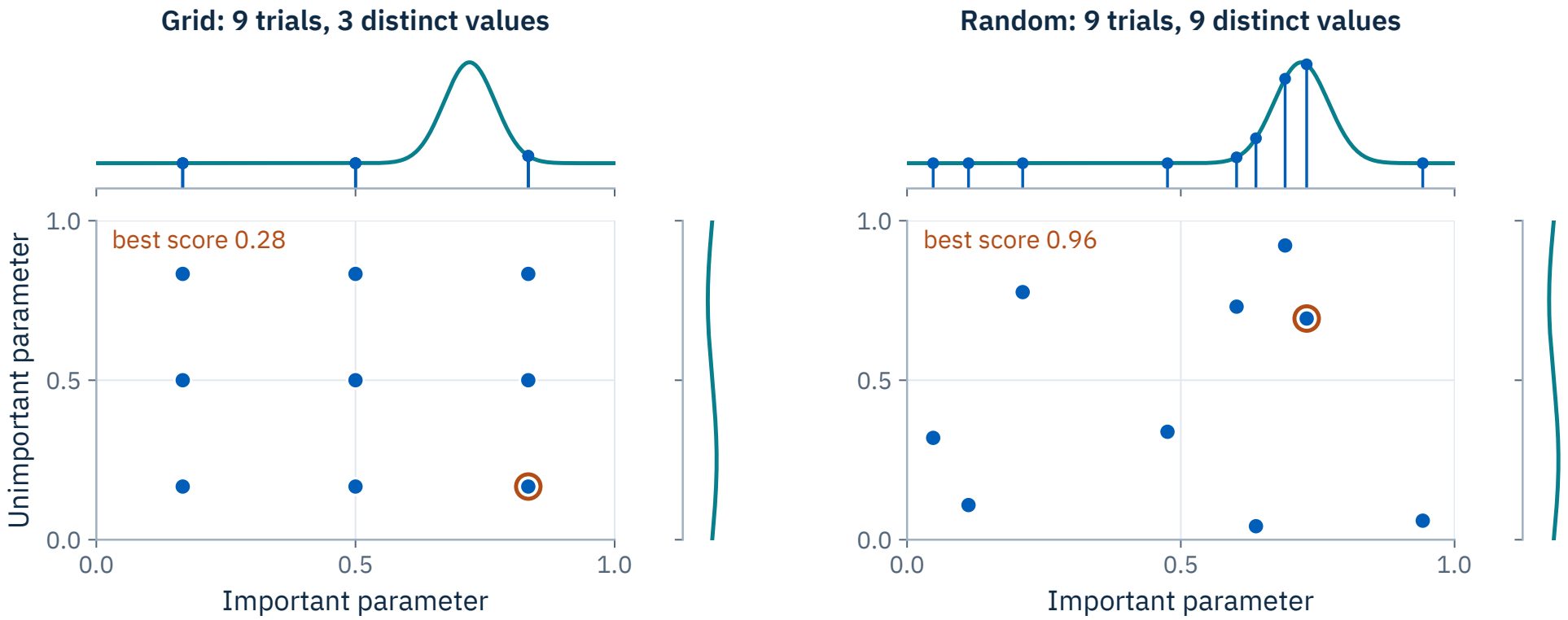
A tuning plan defines the experiment

1. **Fix the question** — target, prediction-time features, and the split policy
2. **Choose the yardstick** — one primary metric plus practical constraints
3. **Bound the search** — candidate space, log scales, compute budget, stopping rule
4. **Keep a ledger** — record every trial, including failures and reruns
5. **Protect the test** — the final partition is used once, at the end

◆ The ledger shows how many candidates were tried, which is what the winner's score must be judged against.

Random search covers the parameters that matter

With 9 trials, a 3 × 3 grid tries 3 values of the important parameter; random search tries 9.



Synthetic objective with a narrow optimum on the important axis (teal curves). One random draw (seed 40); 9 random trials land within 0.05 of the optimum with probability 1 − 0.9 to the 9th power = 0.61.

Random search explores a bounded space

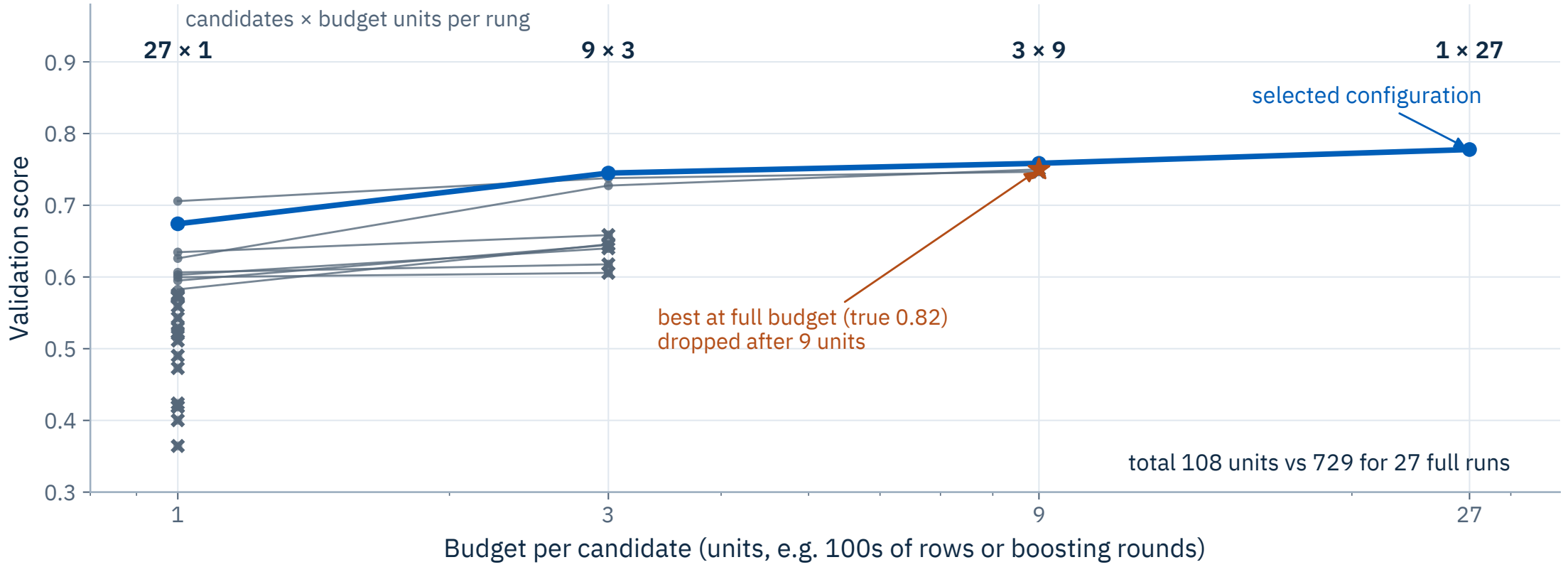
```
from scipy.stats import loguniform
from sklearn.model_selection import RandomizedSearchCV

search = RandomizedSearchCV(
    model, {"model__C": loguniform(1e-3, 1e2)},
    n_iter=12, scoring="average_precision",
    cv=cv, random_state=42, n_jobs=-1
)
search.fit(X_train, y_train)
```

Pass groups to fit when cv requires asset or well groups.

Successive halving spends budget on survivors

Start many candidates cheaply, keep the best third, and triple the budget of the survivors.

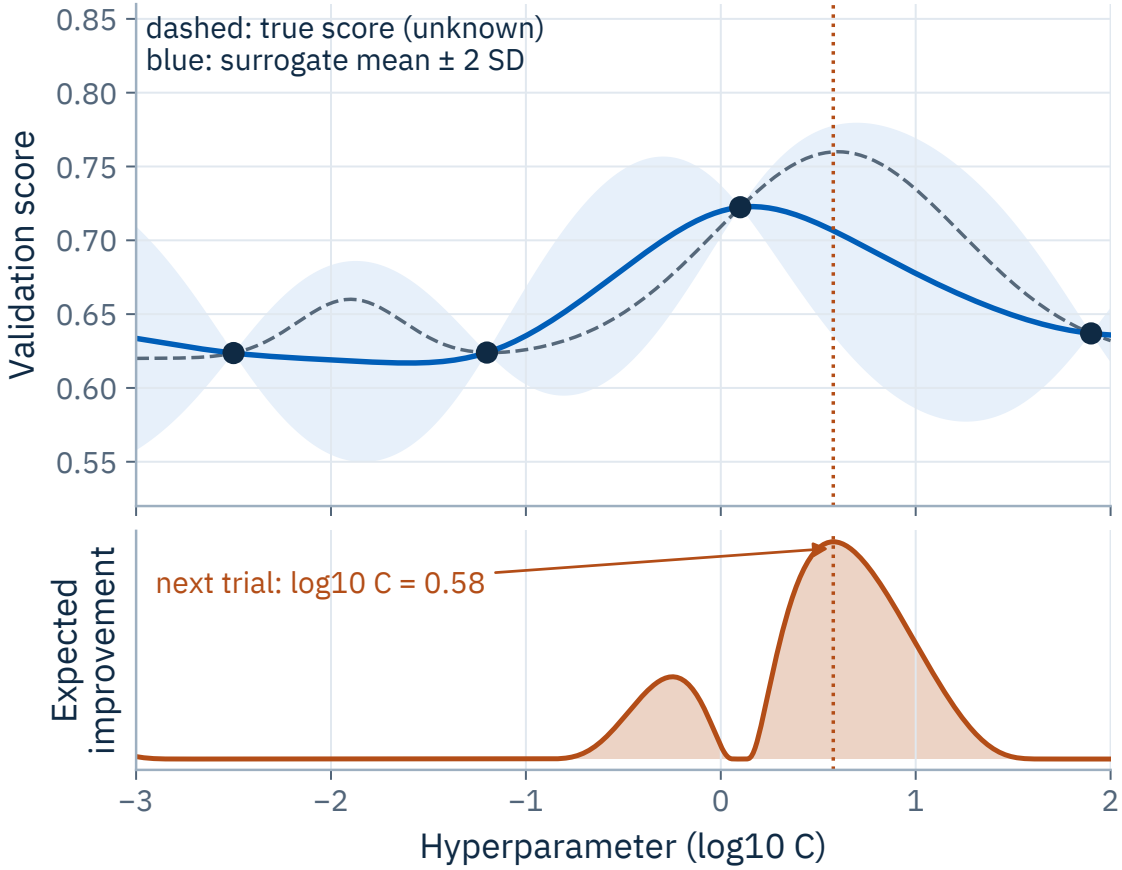


Simulated learning curves for 27 candidates (seed 8); the top third is kept at each rung (factor 3).

Bayesian optimization uses earlier trial results

$$\mathrm{EI}(x) = E[\max(f(x) - f^{+}, 0)]$$

- A surrogate estimates performance across the search space
- The acquisition rule balances promising and uncertain settings
- Each completed trial updates the next choice
- Keep the same validation design; record the whole budget



Synthetic 1-D objective; scikit-learn GaussianProcessRegressor (Matern kernel, fixed hyperparameters) after four trials.

The best of many noisy scores is biased upward

50 candidate settings, all truly equal, each scored with CV noise of SD 0.02:

0.600

true AP of every candidate

0.645

expected best observed AP

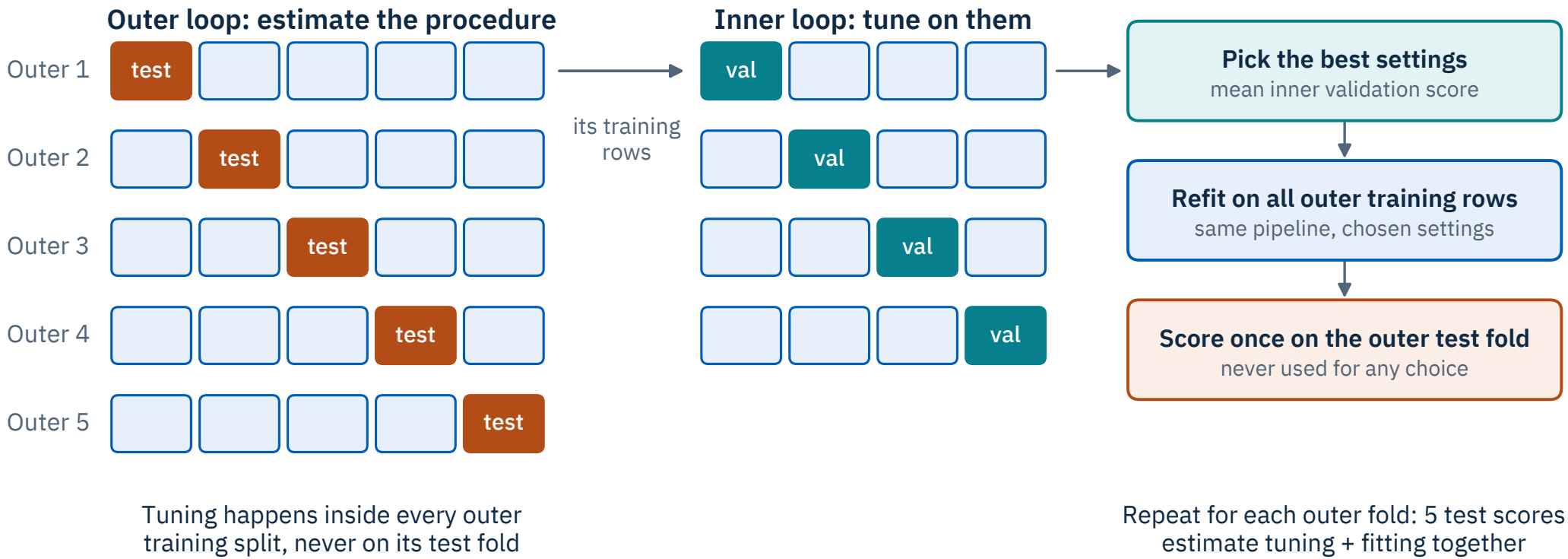
+0.045

optimism from selection alone

◆ Report the chosen model's score from data that played no part in choosing it.

Nested cross-validation scores the selection procedure

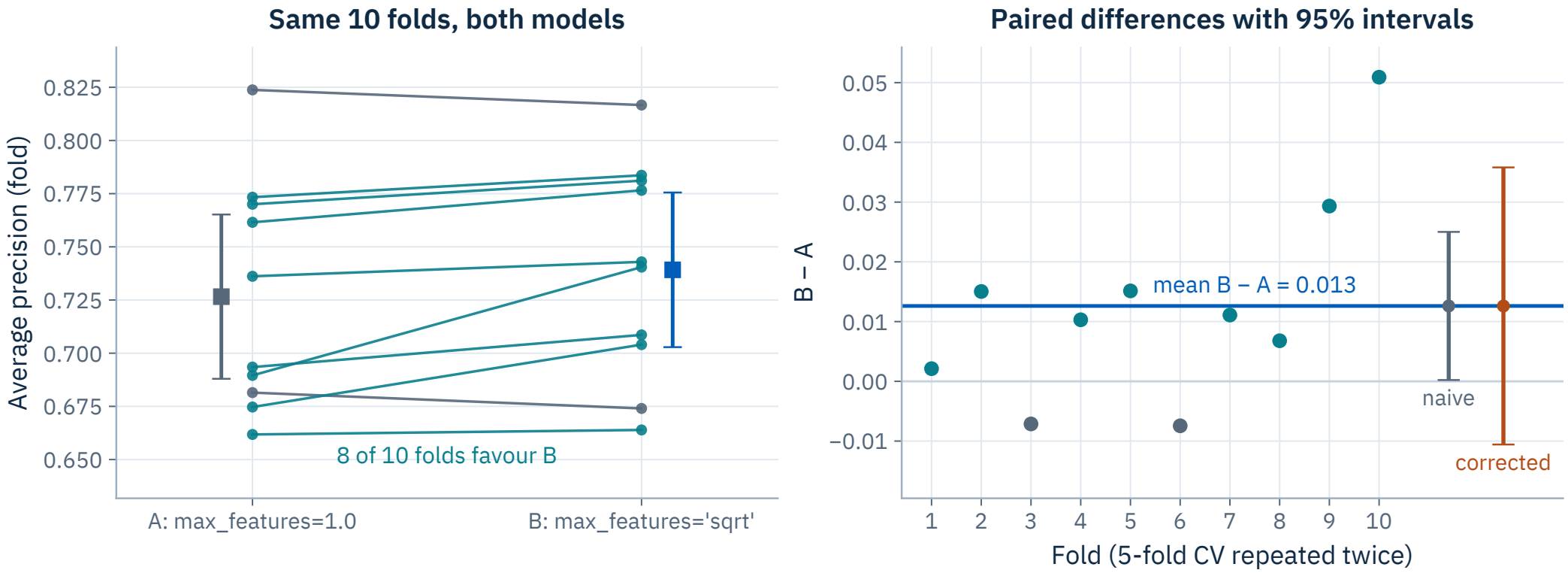
The outer score estimates how well tuning plus fitting generalizes, not one fixed model.



Schematic: five outer folds; inside each outer training split, four inner folds choose the settings.

Compare models on paired folds, with a corrected test

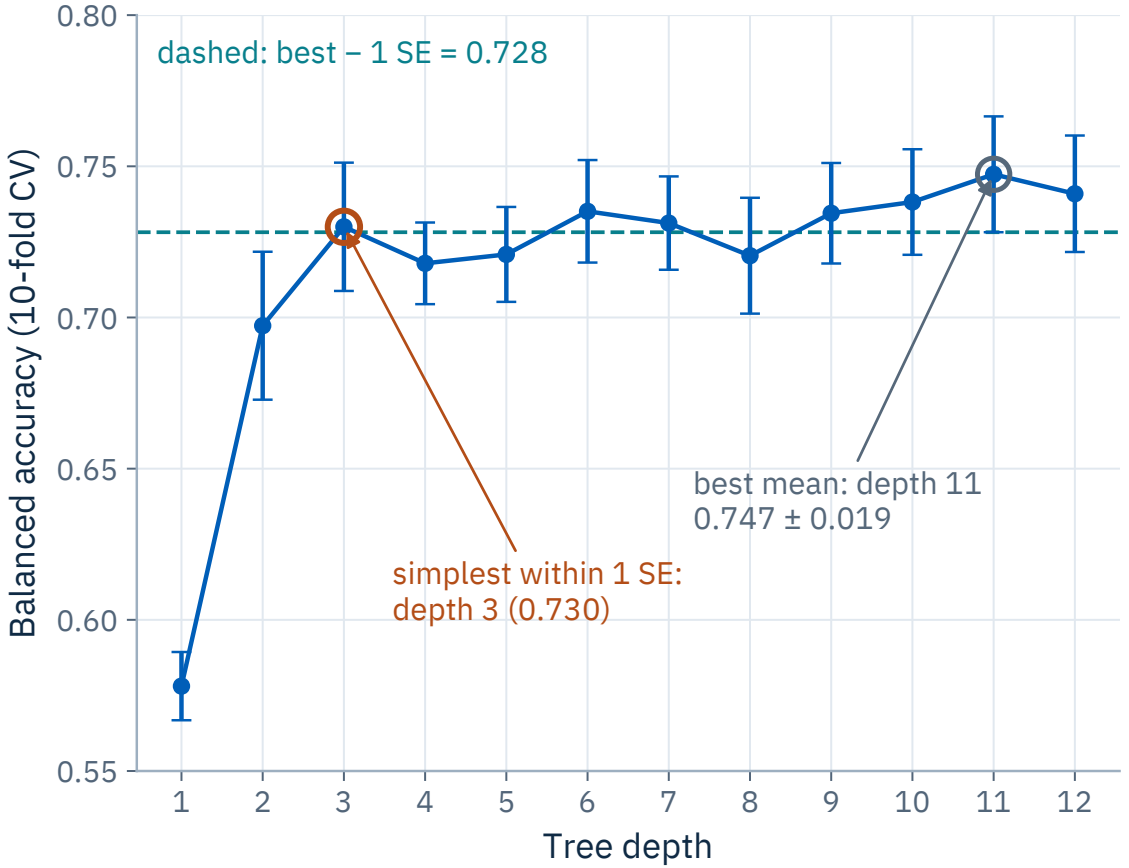
The means' error bars overlap, yet B beats A in 8 of 10 paired folds.



Synthetic data (seed 4, 1,200 rows); RepeatedStratifiedKFold 5 × 2; two random forests differing only in max_features. Corrected interval: Nadeau–Bengio variance, test/train ratio 1/4.

Prefer the simplest model within one standard error

- Find the best mean score and its standard error
- Choose the simplest model within one SE of that best
- Here depth 3 (0.730) instead of depth 11 (0.747)
- A heuristic: fold scores are correlated, so the SE is approximate



Synthetic data (seed 20, 800 rows); DecisionTreeClassifier; 10-fold stratified CV; bars show ±1 standard error.

PRACTICE

8 min · Pairs

Read a tuning log and choose a model

Fold	Depth 3	Depth 6	Depth 12
1	0.61	0.66	0.67
2	0.58	0.62	0.61
3	0.64	0.69	0.70
4	0.55	0.60	0.62
5	0.62	0.68	0.70

1. Which depth has the best mean average precision?
2. Apply the one-SE rule (the SE of depth 12 is 0.019).
3. Is depth 12 clearly better than depth 6 on paired folds?

DISCUSSION

One standard error favours depth 6 in the log

0.660

best mean (depth 12)

0.641

best mean minus one SE

depth 6

simplest within one SE (0.650)

t = 1.8

depth 12 vs 6, naive paired

◆ A 0.01 gain in 4 of 5 folds is weak evidence; decide on cost and simplicity.

PART 6 OF 7

06

Deployment, drift, and monitoring

A validated model is a hypothesis about the future.

- Features that exist at prediction time
- Data drift versus concept drift
- PSI and KS for a shifted input
- The lifecycle loop and a monitoring case

PRACTICE

4 min · Pairs

Feature timing determines whether a forecast is real

A 30-minute warning model uses a centred 30-minute rolling mean, which includes the next 15 minutes.

1. Is that feature available at the forecast origin?
2. How would you repair the feature and the validation design?

Energy features must exist at prediction time

Well logs

- Unit checks and valid transforms
- Within-well gradients, missingness flags

Sensors

- Trailing variability, rates of change
- Operating-state indicators

Production

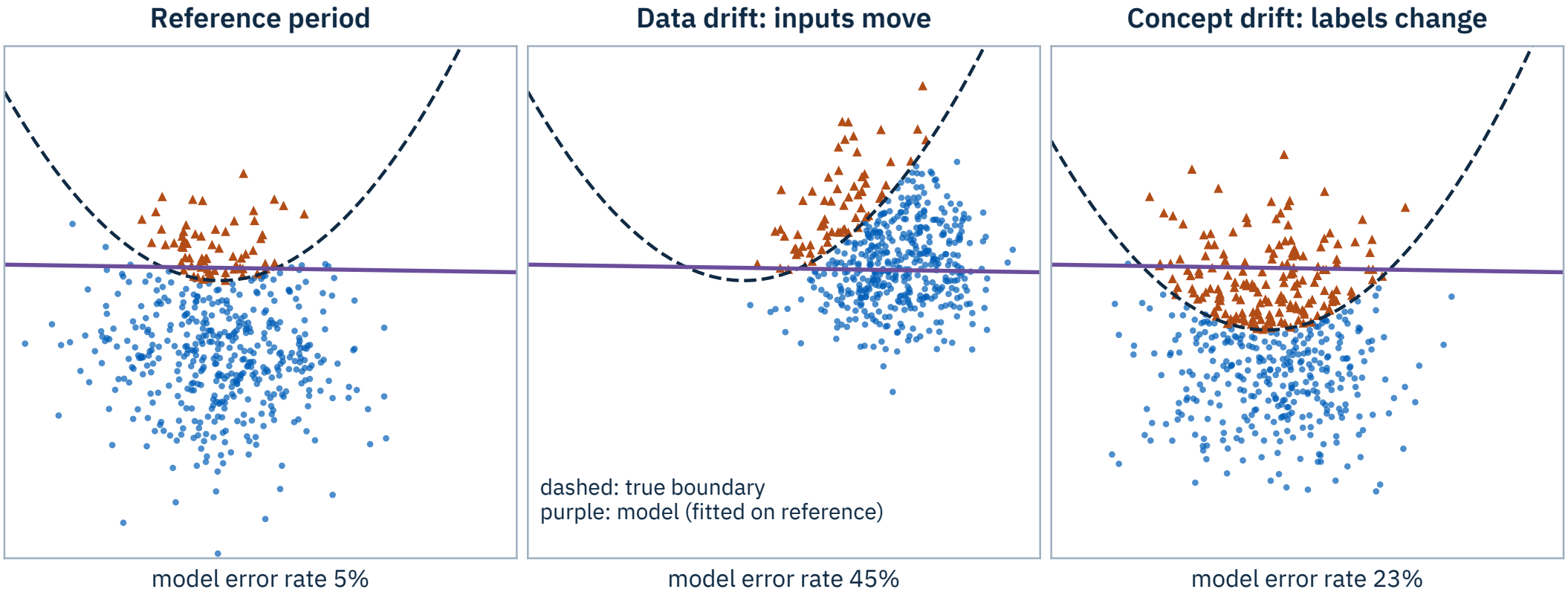
- Lagged rates, downtime indicators
- Covariates known at forecast time

Validation

- Test new features on the intended holdout
- Never carry windows across asset IDs

Data drift and concept drift need different responses

Data drift: P(x) moves. Concept drift: P(y | x) changes. Either breaks a validated model.

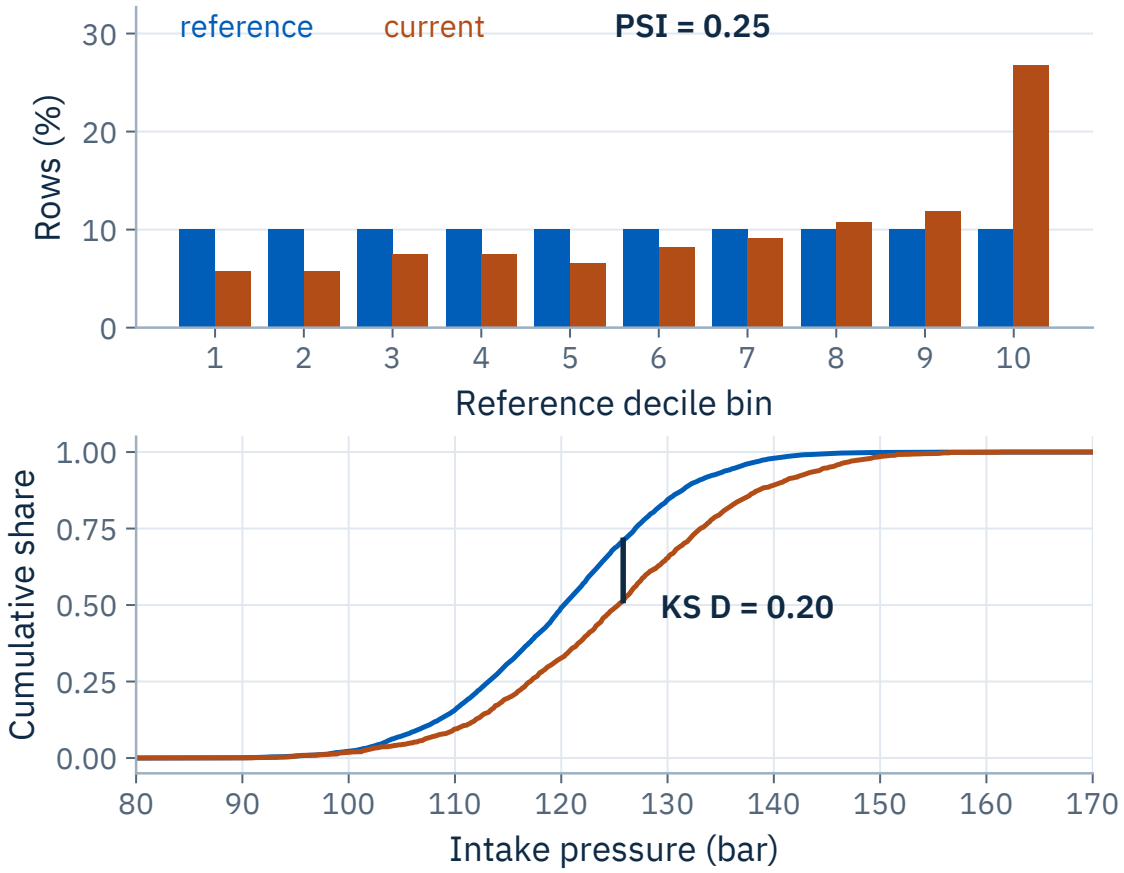


Synthetic data (seed 14): events when x2 > 0.25 x1² + c; logistic regression fitted on the reference period only. Error rates on 500 rows per panel.

PSI and KS quantify a shifted input

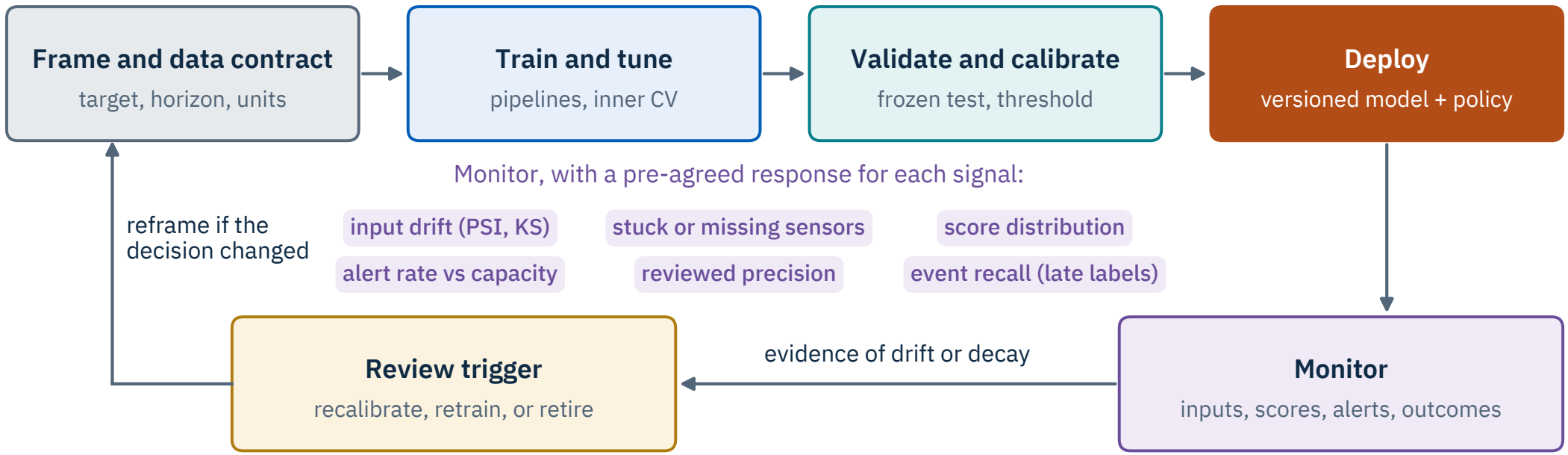
$$\mathrm{PSI} = \sum_i (a_i - e_i) \ln \frac{a_i}{e_i}$$

- e_i and a_i: reference and current shares in bin i
- Here PSI = 0.25 and KS D = 0.20 for a 5-bar shift
- With 1,500 rows the KS p-value is below 10^{-39}
- Set alert levels from PSI between past stable periods



Synthetic intake pressure (seed 15): reference N(120, 10), 5,000 rows; current N(125, 12), 1,500 rows.

Monitoring closes the model lifecycle loop



Schematic model lifecycle with a monitoring feedback loop.

A monitoring case is an experiment, not a promise

Proposed use	Baseline	Evaluation question
ESP monitoring	Rules or logistic classifier	Event recall at a manageable alert rate
Pipeline investigation	Flow-balance residual	False alerts per unit exposure
Well-log quality review	Physical ranges and robust scores	Expert-reviewed ranking quality
Wind turbine performance	Binned power-curve residual	Share of alerts confirmed on inspection
Solar inverter monitoring	Comparison with neighbouring inverters	Confirmed losses per review hour

◆ Each row is a hypothesis to test on your own data; no field performance is claimed.

PART 7 OF 7

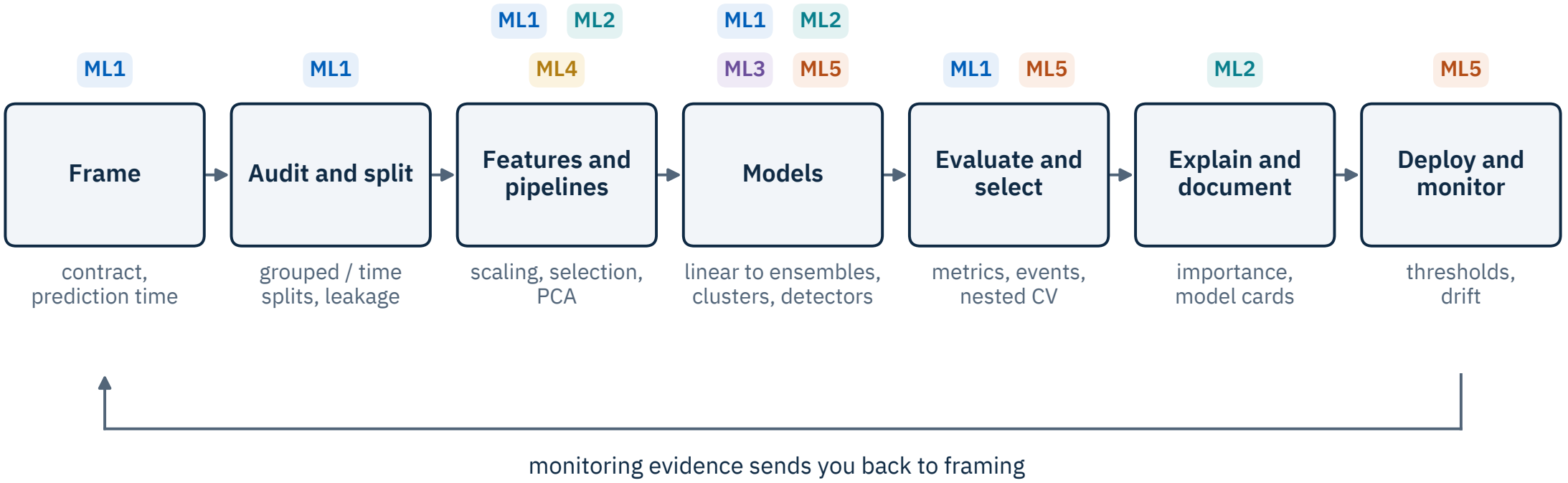
07

Course synthesis

Five modules, one workflow: from a question to a monitored decision.

- The end-to-end workflow and where each module fits
- A synthesis case
- A final review across Modules 1 to 5
- Key takeaways and the exit check

Five modules form one workflow



Schematic of the course workflow; chips show which module teaches each stage.

PRACTICE

4 min · Pairs

Course synthesis: plan a new well-data project

You receive a new well dataset and a request for a high-accuracy model.

1. State the first decision you must clarify.
2. Describe the baseline and the evaluation design.
3. Name the evidence needed before recommending deployment.

PRACTICE

5 min · Groups

Final review: find the flaw in each plan

1. Rows from 40 wells are split at random; the test R^2 is 0.93.
2. A scaler and PCA are fitted on all rows before cross-validation.
3. k-means runs on raw GR (API) and RHOB (g/cc).
4. SMOTE is applied before the train/test split; recall is 0.95.
5. The alert threshold is tuned on the test set that reports precision.

Key takeaways

- ✓ An anomaly score ranks rows; an alert policy turns ranks into decisions
- ✓ Robust baselines and physical rules come before complex detectors
- ✓ Evaluate rare events at the event level and at the real prevalence
- ✓ Rebalancing changes probabilities: recalibrate before using costs
- ✓ Tune inside the validation design; nested CV scores the procedure
- ✓ Monitor inputs, outputs, and outcomes, with the response agreed in advance

PRACTICE

3 min | Individual

Exit check

1. Why can a detector with 80% recall and a 5% false-positive rate produce mostly false alerts?
2. What does nested cross-validation estimate that a tuned CV score does not?
3. Name one monitoring signal and the action it should trigger.

References and further study

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